

Final-Term Papers Solved MCQS with Reference (1 to 45 lectures)

1. _____ convert the relocatable machine code into absolute machine code by linking library and relocatable object files.

   - Assembler
   - **Loader/link-editor**
   - Compiler
   - Preprocessor

2. Parsers take ____ as input from lexical analyzer.

   - Linker
   - **Token**
   - Instruction
   - None of the given

3. The regular expression _____denotes, the set of all strings of a's and b's of length two

- a*
- (a*|b*)*
- (a*b*)*
- **(a|b)(a|b)**

4. _____ is a regular expression for the set of all strings over the alphabet {a} that has an even number of a's.

- aa*
- **(aa)***
- aa*a
- a(aa)*

5. _____ Phase supports macro substitution and conditional compilation.

- Semantic
- Syntax
- **Preprocessing** **Click Here For More Detail**
- None of given

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6. In LL(1) parsing algorithm, ____ contains a sequence of grammar symbols.

- **Stack** **PG # 62**
- Link List
- Array
- None of the given.

7. Consider the grammar

A → B C D
B → h B | €
C → C g | g | C h | i

- **h, g, i**
- g
- h
- None of the given.

8. _____ parsers never shifts into an error state.

- LS
- LT
- **LR**
- LP

9. In parser, the two LL stand for _____ .

- Left-to-right scan of input
- left-most derivation
- **Left-to-right scan of input and left-most derivation** **PG # 54**
- None of the given

10. _____ is elaborated to produce bindings.

- **Declaration** Click Here For More Detail
- Expression
- Command
- None of the given

11. A lexical analyzer generated by ____ is essentially a FSA.

- Dex
- Mex
- Fex
- **Lex**

12. A lexical analyzer generated by **lex** is essentially a PDA (Push Down Automaton).

- True
- **False**

13. The actions (shift, reduce) in a SLR(1) parser depend on a look ahead symbol (_____) .

- **Current input token**
- Next Input Token
- Previous output Token
- Previous Input Token.

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14. The following grammar contains a _____ conflict. S → *A* | *xb*

- **Shift-Reduce**
- First-Reduce
- Shift-First
- Reduce-Reduce

15. S → A | xb

A → aAb | x

This grammar contains a _____ conflict.

- **Shift-Reduce**
- First-Reduce
- Shift-First
- Reduce-Reduce

16. Considering the following grammar:

S → AB
A → € | aA
B → b | bB

- 1
- **2**
- **3**
- 4

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

17. Simple code generation considers one AST node at the time. When the target is a *stack* machine, the code can be generated in one _____ traversal of the AST.

- **Depth-first**
- Breadth-first
- First-first
- Shift-Reduce

18. _____ is a register allocation technique that *always* finds the minimal number of registers needed for a procedure.

- Dangling reference
- **Graph coloring**
- Left Factoring
- Right Recursion

19. **Graph coloring** is a register allocation technique that operates at *individual* basic blocks.

- True
- **False**

20. **Graph coloring** is a register allocation heuristic that *usually* finds the minimal number of registers needed for a procedure.

- **True**
- False

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21. S → a S | Sa | c

This grammar is ambiguous.

- **True**
- False

22. When generating code at the basic block level, the dependency graph must be converted to target code. By identifying _____, instruction selection and instruction ordering can be performed efficiently in a single pass.

- **Ladder sequences**
- Physical sequences
- Logical sequences
- Token sequences

23. _____ can be considered a small compiler since it transforms a source language (assembly) into a less abstract target language (binary object code)

- Parser
- Assembler
- **Lexical analyzer**
- Scanner

24. When memory allocator operates on chunks which include some administrative part and a block of user data. The administrative part includes _____ flag for marking the chunk as free or in-use.

- **One**
- Two
- Three
- Four

25. A parser transforms a stream of tokens into an _____.

- **AST**
- IST
- EST
- ATS

26. The parser generator yacc can handle _____ grammars

- **LL(1)**
- LT(1)
- LS(1)
- LF(1)

27. The parser generator **yacc** can handle LL(1) grammars.

- **True**
- False

28. The **yacc** parser generator can handle LALR(1) grammars.

- **True**
- False

29. Simple code generation considers one AST node at a time. If the target is a *register* machine, the code can be generated in one _____ traversal of the AST, possibly introducing temporaries when running out of registers.

- **Depth-first**
- Breadth-first
- Depth-second
- Breadth-second

30. Attributes whose values are defined in terms of a node's own attributes, node's siblings and node's parent are called _____.

- **Inherited attributes** **PG # 100**
- Physical attributes
- Logical attributes
- Un-synthesized attributes

31. A linker combines multiple object files into a _____ executable object.

- **Single**
- Double
- Triple
- Quadruple

32. The notation _____ instructs YACC to push a computed attribute value on the stack.

- **$$** **PG # 106**
- &&
- ##
- --

33. The following two items

A → P • Q

B → P • Q

can co-exist in an _____ item set

- **LR**
- LS
- LT
- PR

34. When generating a lexical analyzer from a _____ description, the item sets (states) are constructed by two types of "moves": character moves and € moves.

- Character
- Grammar
- **Token**
- Sentence

35. Hybrid IRs combine elements of _____.

- Graphical (structural)
- Linear IRs
- **Both graphical and linear IRs** **PG # 108**
- Non-Linear IRs

36. x[i] = y This is _____ .

- Prefix assignment
- Postfix assignment
- **Index assignment** **PG # 115**
- Non-Index assignment

37. A lexical analyzer generator automatically constructs a _____ that recognizes tokens.

- **FA** **PG # 18**
- PDA
- DP
- Unidirectional Graph

38. if x relop y goto L Above statement is _____

- Abstract jump
- **Conditional jump** **PG # 115**
- While loop
- Unconditional jump

39. In a CFG (Context Free Grammar) the set of terminal and non-terminal symbols must be _____.

- **Disjoint**
- Logical
- Relational
- Joint

40. S → a | B

B → Bb | €

The non-terminal B is left recursive.

- **True**
- False

41. YACC contains built-in support for handling ambiguous grammars resulting in _____ conflicts.

- **Shift-reduce**
- Shift-Shift
- Reduce-reduce
- Reduce-Shift

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42. The _____ translation statements can be conveniently specified in YACC.

- **Syntax-directed** **PG # 128**
- Image-directed
- Sign-directed
- Segment-directed

43. When constructing an LR(1) parser we record for each item exactly in which context it appears, which resolves many conflicts present in _____ parsers based on FOLLOW sets.

- **SLR(1)**
- LRS(1)
- RLS(1)
- SLL(1)

44. Code generation module has to tackle _____.

- Memory management
- Instruction selection
- Instruction scheduling
- **All of the given** **PG # 129**

The code generation module has to tackle a number of issues. Memory management, Instruction selection, Instruction scheduling, Register allocation

45. For convenience, lexical analyzers should read the complete _____ program into memory.

- **Input**
- Output
- Input and output
- Tokens

46. Considering the following grammar:

S → A | x
A → aAb | x

The grammar contains a _____ conflict.

- **Reduce-reduce**
- First-first
- Shift-shift
- Shift-reduce

47. SLR (1) parsers only reduce a production rule when the current input token is an element of the FOLLOW set of that rule.

S → A B

A → € | aA

B → b | bB

- FOLLOW (A) contains 2 elements.

- True
- **False**

48. SLR (1) parsers only reduce a production rule when the current input token is an element of the FOLLOW set of that rule.

S → A B

A → a | aA

B → € | bB

- FOLLOW (A) contains 2 elements.

- **True**
- False

49. The order in which the DAG is traversed can lead to _____ code

- **Better** **PG # 143**
- Worse
- Large
- Garbage

50. Register allocation problem uses the strategy of _____.

- **Graph coloring** **PG # 144**
- Graph nodding
- Graph edging
- Graph patching

51. Typical compilation means programs written in high-level languages to low-level _____.

- **Object code** **PG # 06**
- Byted code
- Unicode
- Object code and byte code

52. In compilation process, Hierarchical analysis is also called _____.

- Parsing
- **Syntax analysis.** **Click here for more detail**
- Parsing and syntax analysis
- None of the given

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53. IR (Intermediate Representation) stores the value of its operand in _____.

- **Registers** **PG # 10**
- Memory
- Hard disk
- None of the given

54. A lexeme is a sequence of characters in the source program that is matched by the pattern for a _____.

- Linker
- **Token**
- Control flow
- None of the given

55. Parsers take _____ as input from lexical analyzer.

- Linker
- **Token**
- Instruction
- None of the given

56. What kind of abstract machine can recognize strings in a regular set?

- **DFA**
- NFA
- PDA
- None of the given

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57. In multi pass compiler, during the first pass it gathers information about _____.

- **Declaration**
- Bindings
- Static information
- None of the given

58. In DFA minimization, we construct one _____ for each group of states from the initial DFA.

- **State** **PG # 30**
- NFA
- PDA
- None of the given

59. _____ (Lexical Analyzer generator), is written in java.

- Flex
- **Jlex** **PG # 31**
- Complex
- None of the given

60. In Flex specification file, different sections are separated by _____.

- **%%** **PG # 31**
- &&
- ##
- None of the given

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61. Recursive _____ parsing is done for LL(1) grammar.

- **Decent**
- Ascent
- Forward
- None of the given

62. Alternative of the backtrack in parser is Look ahead symbol in _____.

- **Input**
- Output
- Input and output
- None of the given

63. Parser takes tokens from scanner and tries to generate _____.

- Binary search tree
- **Parse tree**
- Binary search tree and parse tree.
- None of the given

64. In predictive parsing table, the rows represents _____.

- Terminals
- Both non-terminal and terminal
- **Non-terminal**
- None of the given

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65. A **predictive parser** is a **top-down** parser.

- **True**
- False

66. In LL(1) parsing algorithm, _____ contains a sequence of grammar symbols.

- **Stack** **PG # 62**
- Link list
- Array
- None of the given

67. Bottom-up parsing uses only _____ kinds of actions.

- **Two** **PG # 71**
- Three
- Four
- Five

Bottom-up parsing uses only two kinds of actions: 1. Shift, 2. Reduce

68. Bottom-up parsers handle a _____ class grammars.

- **Large** **PG # 49**
- Small
- Medium
- None of the given

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69. The shift action _____ a terminal on the stack.

- **Pushes** **PG # 73**
- Pops
- Both push and pops
- None of the given

70. Reduce action _____ zero or more symbols from the stack.

- Pushes
- **Pops** **PG # 73**
- Both push and pops
- None of the given

71. In compilers, linear analysis is also called _____.

- Lexical analysis
- Scanning
- **Lexical analysis and scanning**
- None of the given

72. Back End of two-pass compiler uses _____ algorithm.

- O(n)
- O(n log n)
- **NP Complete**
- None of the given

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73. The Back End of a compiler consist of _____.

- **Instruction selection**
- Register allocation
- Instruction scheduling
- All of the given

74. In Back End module of compiler, optimal register allocation uses _____.

- O(log n)
- O(n log n)
- **NP-Complete**
- None of the given

75. A lexeme is a sequence of characters in the source program that is matched by the pattern for a _____.

- Linker
- **Token**
- Control flow
- None of the given

76. _____ is a regular expression for the set of all strings over the alphabets **{a}** that has an even number of **a**'s.

- aa*
- (aa)*
- **aa*a**
- a(aa)*

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77. _____algorithm is used in DFA minimization.

- James's
- Robert's
- **Hopcroft's** **PG # 25**
- None of the given

78. _____ is an important component of semantic analysis.

- Code checking
- **Type checking** **PG # 39**
- Flush checking
- None of the given

79. In _____, certain checks are performed to ensure that components of a program fit together meaningfully.

- Linear analysis
- Hierarchical analysis
- **Semantic analysis**
- None of the given

80. _____ read the input character and produce sequence of tokens as output.

- **Lexical analyzer**
- Parser
- Symbol table
- None of the given

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

81. _____ of a two-pass compiler is consist of instruction selection, Register allocation and instruction scheduling.

- **Backend**
- Frontend
- Start
- None of the given

82. _____ is evaluated to yield a value.

- Command
- **Expression**
- Declaration
- None of the given

83. A parser transforms a stream of tokens into an AST (Abstract Syntax Tree).

- **True**
- false

84. A parser transforms a stream of characters into a stream of tokens.

- True
- **False**

85. A lexical analyzer transforms a stream of characters into a stream of tokens.

- **True**
- False

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86. S → a | A

A → Aa | a

This grammar is ambiguous.

- **True**
- False

87. The regular expressions **(a+|b)?** and **a+|b?** describe the same set of strings.

- **True**
- False

88. The regular expressions **a*|b*** and **(a|b)*** describe the same set of strings.

- True
- **False**

89. The regular expressions **a+a** and **a*aa** describe the same set of strings.

- **True**
- False

90. A lexical analyzer *generator* automatically construct a FSA (Finite State Automaton) that recognizes tokens.

- The generator is driven by a **regular description**

- **True**
- False

91. The transition table in a lexical analyzer records for each state (row) which token, if any, is recognized in that state. - For each token there may be more than one "recognizing" row in the table.

- **True**
- False

92. A **recursive descent** parser is based on a PDA (Push Down Automaton).

- **True**
- False

93. A **bottom-up** parser creates the nodes in the AST in pre-order.

- True
- **False**

94. A **top-down** parser creates the nodes in the AST (Abstract Syntax Tree) in preorder.

- **True**
- False

95. A _____ parser creates the nodes in the AST in preorder.

- **Top – Down**
- Bottom – Up
- Middle – Ware
- Straight

96. The stack used in a bottom-up parser contains an alternating sequence of states and grammar symbols.

- **True**
- False

97. The following two items

A→ P • Q

B → P Q •

Can coexist in an LR item set.

- True
- **False**

98. The following two items

A → x • B

B → • y

Can coexist in an LR item set.

- **True**
- False

99. The following two items

A → P • P

B → Q • Q

Can coexist in an LR item set.

- True
- **False**

100. S → A | xb

A → aAb | x

This is an LALR(1) grammar.

- **True**
- False

101. A **linker** combines multiple object files into a single executable object.

- **True**
- False

102. **Data-flow equations** can be solved efficiently by using bitwise boolean instructions (AND, OR, etc.).

- **True**
- False

103. **Data-flow equations** operate with IN, OUT, GEN, and KILL sets.

- **True**
- False

104. When **threading** an AST it might be necessary to introduce additional (join) nodes to ensure that each language construct has a single exit point.

- **True**
- False

105. An **iterative** interpreter operates on a threaded AST.

- **True**
- False

106. S → A | B

A → € | aA

B → b | bB

FIRST(S) contains _____ elements.

- 2
- **3**
- 4
- None

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107. The following set

S → • A x {$}

A → • a {x}

A→ • a A {x}

is a valid LR(1) item set

- **True**
- False

108. S → Ab

A → Aa | €

FIRST(S) contains 3 elements.

- True
- **False**

109. The regular expressions **a(b|c)** and **ab|ac** describe the same set of strings.

- **True**
- False

110. S → a | B

B → Bb | E

The non-terminal _____ is left recursive.

- **B**
- a
- E
- None of the given

111. In PASCAL _____ represent the inequality test.

- :=
- =
- **<>**
- None of the given

112. In parser the two LL stand(s) for _____.

- Left-to-right scan of input
- left-most derivation
- **All of the given**
- None of the given

113. Consider the grammar

A →B C D
B → h B | epsilon
C → C g | g | C h | i
D → A B | epsilon

First of C is _____.

- **g, I look down for reference**
- g
- h , i
- i

خوبصورتی علم و ادب سے ہوتی ہے لباس و حسن سے نہیں

114. Three-address codes are often implemented as a _____.

- **Set of quadruples** **PG # 104**
- Set of doubles
- Set of Singles
- None of the given

115. What does following statement represent? x[i] = y

- Prefix assignment
- Postfix assignment
- **indexed assignment** **PG #107**
- None of the given

116. _____ convert the reloadable machine code into absolute machine code by linking library and reloadable object files.

- Assembler
- **Loader/link-editor**
- Compiler
- Preprocessor



117. Consider the following grammar,

A → B C D

B → h B | episilon

C → C g | g | C h | i

D → A B | episilon

First of A is _____.

- **h, g, i**
- g
- h
- None of the given

118. One of the core tasks of compiler is to generate fast and compact executable code.

- **True** **PG # 14**
- False

119. Compilers are sometimes classified as.

- Single pass
- Multi pass
- Load and go
- **All of the given** **Click here for more detail**

120. In multi pass compiler during the first pass it gathers information about _____.

- **Declaration** **Click here for more detail**
- Bindings
- Static information
- None of the given

121. We can get an LL(1) grammar by _____ .

- Removing left recurrence
- Applying left factoring
- **Removing left recurrence and Applying left factoring** **Click for more detail**
- None of the given

122. Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string "abbcde" can be parsed bottom-up by the following reduction steps:

(i) aTbcde
(ii) aTde
(iii) aTUe
(iv) S

So, what can be a handle from the following?

- **The whole string, (aTUe)** **PG # 68**
- The whole string, (aTbcde)
- The whole string, (aTde)
- None of the given

123. When generating a lexical analyzer from a token description, the item sets (states) are constructed by two types of "moves": character moves and ____ moves.

- **E (empty string)** **P # 18**
- #
- @
- none of given

124. Which of the following statement is true about Two pass compiler.

- Front End depends upon Back End
- **Back End depends upon Frond End** **PG # 5**
- Both are independent of each other
- None of the given

125. _____avoid hardware stalls and interlocks.

- Register allocation
- **Instruction scheduling** **PG #10**
- Instruction selection
- None of given

126. Front end of two pass compiler takes_____ as input.

- **Source code** **PG # 5**
- Intermediate Representation (IR)
- Machine Code
- None of the Given

127. In Three-pass compiler _____ is used for code improvement or optimization.

- Front End
- **Middle End** **PG # 10**
- Back End
- Both Front end and Back end

128. _____ of a two-pass compiler is consists of Instruction selection, Register allocation and Instruction scheduling.

- **Back end** **PG # 9**
- Front end
- Start
- None of given

129. NFA is easy to implement as compared to DFA.

- True
- **False** **PG # 19**

130. In a transition table cells of the table contain the _____ state.

- Reject state
- **Next state** **PG #18**
- Previous state
- None of the given

131. The regular expressions a*|b* and (a|b)* describe the _____set of strings.

- Same
- **Different**
- Onto

132. A canonical collection of sets of items for an augmented grammar, C is constructed as _____

- For each set / in C and each grammar symbol X where goto (C, X) is empty and not in C add the set goto (C, X) to C.
- **The first set in C is the closure of {[S' --> .S]}, where S' is starting symbol of original grammar and S is the starting non-terminal of augmented grammar.** **PG # 72**
- The first set in C is the closure of {[S' --> .S]}, where S is starting symbol of original grammar and S' is the Starting non-terminal of original grammar.

133. The _____translation statements can be conveniently specified in YACC

- **Syntax-directed** **PG # 120**
- Image-directed
- Sign-directed
- None of the given.

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134. Attributes whose values are defined in terms of a node's own attributes, node's siblings and node's parent are called _____.

- **Inherited attributes** **PG # 92**
- Physical attributes
- Logical attributes
- Un-synthesized attributes

135. Consider the grammar

A → B C D

B → h B | epsilon

C → C g | g | C h | i

D → A B | epsilon

Follow of B is _____.

- **h**
- g, h, i, $
- g, i
- g

136. Consider the grammar A --> B C D

B --> h B | epsilon

C --> C g | g | C h | i

D --> A B | epsilon

Follow of C is _____.

- **g, h, i, $** **PG # 47**
- g, h, $
- h, i, $
- h, g, $

137. The test of string is described by a rule called a _____, associated with token.

- Character
- Loader
- **Pattern**
- None of the given

138. Bottom up parsing is also called _____.

- **LR Parsing** **PG # 70**
- LT Parsing
- LS Parsing
- None of the given

Bottom-up parsing is more general than top-down parsing. Bottom-up parsers handle a large class of grammars. It is the preferred method in practice. It is also called *LR* parsing;

139. A DFA can be reconstructed from NFA using the subset construction, similar to one used for_____

- **Lexical Analysis** **PG # 82**
- Physical Analysis
- Logical Analysis
- Parsing

140. Which of the following system software resides in the main memory always?

- Text editor
- Assembler
- Linker
- **Loader**

141. _____ plays an important role in code optimization.

- **DAG** **PG # 143**
- Lexical Analyzer
- AGD
- Memory Management

142. LR parsers can handle _____ grammars.

- **Left-recursive** **PG # 63**
- file-recursive
- End-recursive
- Start-recursive

143. Performing common sub expression elimination on a dependency graph requires the identification of nodes with the same operator and operands. When using a hash table (with a hash function based on operator and operands) all_____ nodes can be identified in linear time.

- **Common**
- Uncommon
- Next
- Previous

144. Linear IRs resembles pseudo-code for same _____.

- Automated Machine
- Mechanical machines
- Token machines
- **Abstract machine** **PG # 100**

145. S → A B

A → € | aA

B → € | bB

- FIRST(S) contains ___ elements

- **3** **PG # 46**

First{S} = {e,a,b}

146. Dotted items (T→ α • β) record which part of a token has already been matched. Integer? ([0-9])+ • this is a _____ item.

- **Reduced**

147. A _____ is a top down parser.

- **Predictive Parsing** **PG # 46**

148. Dotted items (T→ α • β) record which part of a token has already been matched. There are two kinds of basic items: **shift items** and **reduce items**. Integer! (•[0-9])+

- This is a shift item.

- **True**
- False

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

Note: Give me a feedback and your Suggestion also If you find any mistake in mcqz please mail at above mentioned email address. And tell me your answer with references.





please pray for me
and I will pray for you too



THE END

Campus (AKMP01)

ARSLAN ARSHAD
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Solved by:	Sahar (well wisher)	Class	BSCS 6th Semester
Subject	CS606 (COMPILER CONSTRUCTION)	Solution Type:	Final Term Solved MCQZ and Quizez including Papers of Year : 2013,2012,2011,2010,2009… 2006
Institute:	Virtual University of Pakistan		

Final term QUIZEZ
Year 2013

1)

The regular expressions a*|b* and (a|b)* describe the _____set of strings.

Same

Different

Onto

2)

A canonical collection of sets of items for an augmented grammar, C is constructed as -----

For each set / in C and each grammar symbol X where goto (C, X) is empty and not in C add the set goto (C, X) to C.

The first set in C is the closure of {[S' --> .S]}, where S' is starting symbol of original grammar and S is the starting non-terminal of augmented grammar. Page no : 72

The first set in C is the closure of {[S' --> .S]}, where S is starting symbol of original grammar and S' is the starting non-terminal of original grammar.

3)

S --> a | B

B --> Bb | E The non-terminal _____ is left recursive.

Answer: B

4)

Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string "abbcde" can be parsed bottom-up by the following reduction steps:

(i) aTbcde
(ii) aTde
(iii) aTUe
(iv) S So, what can be a handle from the following?

The (a) in (aTUe) not confirm

The (a) in (aTUe)

The (d) in (aTUe)

5)

In an attribute grammar each production rule(N--> a) has a corresponding attribute evaluation rule that describes how to compute the values of the _____attributes of each particular node N in the AST.

Select correct option:

1. **Synthesized page no : 92**
2. Complete
3. Free
4. Bounded

6)

Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string “abbcde” can be parsed bottom-up by the following reduction steps: (i) aTbcde (ii) aTde (iii) aTUe (iv) S So, what can be a handle from the following?

Select correct option:

1. **The whole string, (aTUe) Page no : 68**
2. The whole string, (aTbcde)
3. The whole string, (aTde)
4. None of the given

7)

When constructing an LR(1) parser we record for each item exactly in which context it appears, which resolves many conflicts present in _____parsers based on FOLLOW sets.

Select correct option:

1. SLR(1)
2. LRS(1)
3. RLS(1)
4. **None of the given**

8)

The _____translation statements can be conveniently specified in YACC

Select correct option:

1. **Syntax-directed Page no : 120**
2. Image-directed
3. Sign-directed
4. None of the given.

9)

We use ----- to mark the bottom of the stack and also the right end of the input when considering the Stack implementation of Shift-Reduce Parsing.

Select correct option:

1. Epsilon

2. #
3. **$ Page no : 65**
4. None of the given

10)

Grammars with LL(1) conflicts can be made LL(1) by applying left-factoring, substitution, and left-recursion removal. Left-factoring takes care of _____conflicts.

Select correct option:

1. FIRST/FIRST
2. FIRST/SECOND
3. SECOND/FIRST
4. **None of the given**

11)

When generating a lexical analyzer from a token description, the item sets (states) are constructed by two types of “moves”: character moves and ____ moves.

Select correct option:

1. **E (empty string) Page no : 18**
2. #
3. @
4. none of given

12)

Bottom-up parsers handle a _____ class of grammars.

Select correct option:

1. **large Page no : 63**
2. small
3. medium
4. none of the given

13)

Let a grammar G = (Vn, Vt, P, S) is modified by adding a unit production S’--> S to the grammar and now starting non-terminals becomes S’ and grammar becomes G’ = (Vn U {S’}, Vt, P U{S’ --> S}, S’). The Grammar G’ is called the ----------

Select correct option:

1. **Augmented Grammar Page no : 76**
2. Lesser Grammar
3. Anonymous Grammar
4. none of given

14)

The LR (1) items are used as the states of a finite automaton (FA) that maintains information about the parsing stack and progress of a shift-reduce parser.

Select correct option:

1. **True** **Page no: 74**
2. false

Back patching to translate flow-of-control statements in ____ pass.

Select correct option:

1. **one** **Page no : 111**
2. two
3. three
4. all of the given

15)

Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string “abbcde” can be parsed bottom-up by the following reduction steps: (i) aTbcde (ii) aTde (iii) aTUe (iv) S So what can be a handle from the following?

Select correct option:

1. The second (b) in (abbcde)
2. The first (b) in (abbcde)
3. The substring (cd) in (abbcde)
4. None of the given

16)

S --> A | xb
A --> aAb | x This grammar contains a reduce-reduce conflict.
Select correct option:

True
False

17)

Yacc contains built-in support for handling ambiguous grammars resulting in _____ conflicts.
Select correct option:

Shift-reduce
Shift-Shift
Shift-second
None of the given

18)

A lexical analyzer generator automatically constructs a _____ that recognizes tokens.
Select correct option:

FA
PDA
DP
None of the given

19)

When generating a lexical analyzer from a token description, the item sets (states) are constructed by two types of “moves”: character moves and ____ moves.
Select correct option:

E (empty string) **Repeated**
#
@
None of the given

20)
Attributes whose values are defined in terms of a node’s own attributes, node’s siblings and node’s parent are called _____ .
Select correct option:

Inherited attributes **Page no : 92**
Physical attributes
Logical attributes
Un-synthesized attributes

21)
Register allocation by graph coloring uses a register interference graph. _____ nodes in the graph are joined by an edge when the live ranges of the values they represent overlap.
Select correct option:

Two **page no : 136**
Three
Four
Five

22)
The following two items A -> P • Q B -> P • Q can co-exist in an _____ item set.
Select correct option:

LR
LS
LT
PR

23)

Goto L statement represent

Select correct option:

Unconditional jump **Page no : 107**

24)

Three-address codes are often implemented as a _____.

Select correct option:

Set of quadruples **Page no : 104**

25)

Attributes of a node whose values are defined wholly in terms of attributes of node's children and from constants are called _____.

Synthesized attributes Page no : 92

26)

The error handling mechanism of the yacc parser generator pushes the input stream back when inserting 'missing' tokens.

Select correct option:

Answer: True

27)

The notation _____ instructs YACC to push a computed attribute value on the stack.

Select correct option:

Answer: $$ Page no : 98

28)

Flow of values used to calculate synthesized attributes in the parse tree is:

Select correct option:

Answer: Bottom-up Page no: 92

29)

Following statement represents: if x relop y goto L

Answer: conditional loop Page no : 107

30)

What does following statement represent? x[i] = y

Answer: indexed assignment Page no : 107

31)

S --> A B

A --> e | aA

B --> e | bB - FIRST(S) contains ___ elements

Answer: 3 Page no : 46

First{S}={e,a,b}

32)

Dotted items (T→a •b) record which part of a token has already been matched. Integer? ([0-9])+ • This is a _____ item.

Answer: Extended Page no : 73

33)

Parser generator for the grammar LALR (1) is:

Answer: YACC, Bison, CUP Page no: 88

34)

A _____ is a top down parser.

Answer: Predictive Parsing Page no: 46

35)

A lexical analyzer transforms a stream of tokens. The tokens are stored into symbol table for further processing by the parser.

Answer: True Page no: 99

FINAL TERM QUIZEZ

YEAR 2012+previous

Question # 1 of 10 (Start time: 06:31:04 PM) Total Marks: 1
Parser always gives a tree like structure as output
Select correct option:
True Page no : 8

False

Question # 2 of 10 (Start time: 06:32:19 PM) Total Marks: 1
Lexer and scanner are two different phases of compiler
Select correct option:

True
False Page no : 25

Question # 3 of 10 (Start time: 06:33:49 PM) Total Marks: 1
Intermediate Representation (IR) stores the value of its operand in _____ .
Select correct option:

Registers Page no : 10
Memory
Hard disk
Secondary storage

Question # 4 of 10 (Start time: 06:35:17 PM) Total Marks: 1
In Flex specification file different sections are separated by _____ .
Select correct option:

%% Page no: 26
&&
##
\\

Question # 7 of 10 (Start time: 06:38:21 PM) Total Marks: 1
_____ phase which supports macro substitution and conditional compilation.
Select correct option:

Semantic
Syntax
Preprocessing
None of given

Question # 9 of 10 (Start time: 06:40:30 PM) Total Marks: 1
Flex is an automated tool that is used to get the minimized DFA (scanner).
Select correct option:

True
False Page no: 26

Question # 10 of 10 (Start time: 06:41:58 PM) Total Marks: 1
_____tree in which each node represents an operator and children of the node represent the operands.
Select correct option:

Abstract syntax Page no : 100
Concrete syntax
Parse
None of the given

Question # 2 of 10 (Start time: 06:47:42 PM) Total Marks: 1
In _____certain checks are performed to ensure that components of a program fit together meaningfully.
Select correct option:

Linear analysis
Hierarchical analysis

Semantic analysis **Page no : 33**
None of given

Question # 3 of 10 (Start time: 06:48:58 PM) Total Marks: 1
In compilation process Hierarchical analysis is also called
Select correct option:

Parsing
Syntax analysis
Both Parsing and Syntax analysis
None of given

Question # 5 of 10 (Start time: 06:51:46 PM) Total Marks: 1
Which of the following statement is true about Two pass compiler.
Select correct option:

Front End depends upon Back End
Back End depends upon Frond End **page no : 5**
Both are independent of each other
None of the given

Question # 8 of 10 (Start time: 06:55:41 PM) Total Marks: 1
_____ algorithm is used in DFA minimization.
Select correct option:

James's
Robert's
Hopcroft's **Page no : 19**
None of given

Question # 9 of 10 (Start time: 06:56:23 PM) Total Marks: 1
Ambiguity can easily be handled by Top-down Parser
Select correct option:

True
False

LR parsers can handle _____ grammars.

Left-recursive Page no: 163

file-recursive

End-recursive

Start-recursive

_____ convert the reloadable machine code into absolute machine code by linking library and reloadable object files.

Assembler

Loader/link-editor

Compiler

Preprocessor

Consider the grammar A --> B C D

B --> h B | epsilon

C --> C g | g | C h | i

D --> A B | epsilon

Follow of B is _____ .

h

g, h, i, $

g, i

g

Consider the grammar A --> B C D

B --> h B | epsilon

C --> C g | g | C h | i

D --> A B | epsilon

Follow of C is _____ .

g, h, i, $ Page no : 47

g, h, $

h, i, $

h, g, $

An important component of semantic analysis is _____ .

code checking

type checking page no : 6

flush checking

None of the given

In PASCAL _____ represent the inequality test.

:=

=

<>
None of the given

Lexical Analyzer generator _____ is written in Java.

Flex

Jlex Page no : 26

Complex

None of given

_____avoid hardware stalls and interlocks.

Register allocation

Instruction scheduling Page no : 10

Instruction selection

None of given

Consider the following grammar,

A --> B C D

B --> h B | episilon

C --> C g | g | C h | i

D --> A B | episilon

First of A is _____ .

h, g, i

g

h

None of the given

Recursive _____ parsing is done for LL(1) grammar.

Decent Page no : 47

Ascent

Forward

Backward

One of the core tasks of compiler is to generate fast and compact executable code.

True

False

Left factoring of a grammar is done to save the parser from back tracking.

True Page no:61

False

Responsibility of _____ is to produce fast and compact code.

Instruction selection

Register allocation

Instruction scheduling

None of given Page no: 9

_____ algorithm is used in DFA minimization.

James's

Robert's
Hopcroft's **Page no:25**
None of given

Compilers are sometimes classified as.

Single pass

Multi pass

Load and go

All of the given

In multi pass compiler during the first pass it gathers information about _____ .
Select correct option:
Declaration
Bindings
Static information
None of the given **

Question # 9 of 10 (Start time: 06:40:30 PM) Total Marks: 1
Flex is an automated tool that is used to get the minimized DFA (scanner).
Select correct option:
True
False

In compilation process Hierarchical analysis is also called
Select correct option:
Parsing
Syntax analysis
Both Parsing and Syntax analysis
None of given

For each language to make LL(1) grammar, we take two steps, 1st is removing left recurrence and 2nd is applying fin sequence.

True
False

_____is evaluated to yield a value.

Command
Expression
Declaration
Declaration and Command

LR parsers can handle _____ grammars.
Left-recursive **page no : 63**
file-recursive
End-recursive
Start-recursive

Optimal registers allocation is an NP-hard problem.

True

False **Page no : 10**

Parser takes tokens from scanner and tries to generate _____ .

Binary Search tree

Parse tree

Syntax trace **Page no : 6**

None of the given

Front end of two pass compiler takes_____ as input.

Source code **Page no: 5**

Intermediate Representation (IR)

Machine Code

None of the Given

In DFA minimization we construct one _____ for each group of states from the initial DFA.

State **Page no : 25**

NFA

PDA

None of given

In Three-pass compiler _____ is used for code improvement or optimization.

Front End

Middle End **Page no : 10**

Back End

Both Front end and Back end

_____ of a two-pass compiler is consists of Instruction selection, Register allocation and Instruction scheduling.

Back end **Page no : 9**

Front end

Start

None of given

NFA is easy to implement as compared to DFA.

True

False **Page no : 19**

We can get an LL(1) grammar by _____ .

Removing left recurrence

Applying left factoring

Removing left recurrence and Applying left factoring

None of the given

Parser always gives a tree like structure as output

True **page no : 37**

False

Intermediate Representation (IR) stores the value of its operand in

Registers
Memory
Hard disk
Secondary storage

In Back End module of compiler, optimal register allocation uses_____ .
O(log n)
O(n log n)
N P-Complete Page no : 10
None of the given

CS 606 Quizez

Can a DFA simulate NFA?

Yes

No

Sometimes

Depend upon nfa

_____ phase which supports macro substitution and conditional compilation.

Semantic

Syntax

Preprocessing

None

Which of the statement is true about Regular Languages?

Regular Languages are the most popular for specifying tokens.

Regular Languages are based on simple and useful theory.

Regular Languages are easy to understand.

All of the given

Lexer and scanner are two different phases of compiler

True

False Page no :13

Lexical Analyzer generator _____ is written in Java.

Flex

Jlex Page no :26

Complex

None of the given

In a transition table cells of the table contain the _____ state.

Reject state

Next state **Page no 18**

Previous state

None of the given

The transition graph for an NFA that recognizes the language (a | b)*abb will have following set of states.

{0}

{0,1}

{0,1,2}

{0,1,2,3} **not sure**

Front end of two pass compiler takes_____ as input.

Source code

Intermediate representation

Machine code

None

Functions of Lexical analyzer are?

Removing white space

Removing constants, identifiers and keywords

Removing comments

All of the given

Question # 1 of 10 (Start time: 07:25:59 PM) Total Marks: 1

Front-end of a two pass compiler is consists of Scanner.

Select correct option:

True

False

Question # 2 of 10 (Start time: 07:26:40 PM) Total Marks: 1

LL(1) parsing is called non-predictive parsing.

Select correct option:

True

False

Question # 3 of 10 (Start time: 07:28:09 PM) Total Marks: 1

Recursive _____ parsing is done for LL(1) grammar.

Select correct option:

Backward

Forward

Ascent

Decent

Question # 4 of 10 (Start time: 07:29:35 PM) Total Marks: 1

In predictive parsing table the rows are _____ .

Select correct option:

Non-terminals

Terminals

Both non-terminal and terminals

None of the given

Question # 5 of 10 (Start time: 07:30:38 PM) Total Marks: 1

We can get an LL(1) grammar by _____ .

Select correct option:

Removing left recurrence

Applying left factoring

Removing left recurrence and Applying left factoring

None of the given

Question # 6 of 10 (Start time: 07:31:48 PM) Total Marks: 1

_____ of a two-pass compiler is consists of Instruction selection, Register allocation and Instruction scheduling.

Select correct option:

Backend

Frontend

Start

Question # 7 of 10 (Start time: 07:32:26 PM) Total Marks: 1

Consider the grammar

A --> B C D

B --> h B | epsilon

C --> C g | g | C h | i

D --> A B | epsilon First of C is _____ .

Select correct option:

g, I look down for reference

g

h i

i

Quiz Start Time: 07:25 PM

Time Left 81

sec(s)

Question # 8 of 10 (Start time: 07:33:12 PM) Total Marks: 1

Alternative of the backtrack in parser is Look ahead symbol in _____ .

Select correct option:

input

output

input and output

none

Question # 9 of 10 (Start time: 07:34:09 PM) Total Marks: 1

AST summarizes the grammatical structure with the details of derivations.

Select correct option:

True

False

Quiz Start Time: 07:25 PM

Time Left 81

sec(s)

Question # 10 of 10 (Start time: 07:35:06 PM) Total Marks: 1

One of the core tasks of compiler is to generate fast and compact executable code.

Select correct option:

True

False

Left factoring is enough to make LL1 grammar

True

False

In LL1() parsing algorithm _____ contains a sequence of grammar symbols.

Stack

Link list

Array

None

Final Term MCQ's and Quizzes
CS606- compiler instruction

Question No: 1 (Marks: 1) - Please choose one
If X is a terminal in A--> aX•?, then this transition corresponds to a shift of ____ from input to top of parse stack.

X
A
a
None of the given

Question No: 1 (Marks: 1) - Please choose one
A canonical collection of sets of items for an augmented grammar, C is constructed as -----

The first set in C is the closure of {[S' --> .S]}, where S is starting symbol of original grammar and S' is the starting non-terminal of augmented grammar.
The first set in C is the closure of {[S' --> .S]}, where S is starting symbol of original grammar and S' is the starting non-terminal of original grammar.
The first set in C is the closure of {[S' --> .S]}, where S is starting symbol of original grammar and S is the starting non-terminal of augmented grammar.
None of these

Question No: 1 (Marks: 1) - Please choose one
An ----- does not need to examin the entire stack for a handle, the state symbol on the top of the stack contains all the information it needs.

LR parser
RL parser
BU parser
None of the given

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Question No: 1 (Marks: 1) - Please choose one
Suppose ? begins with symbol X which may be a terminal (token) or non-terminal. The item can be written as A? Xa•?.

True
False

Question No: 1 (Marks: 1) - Please choose one
YACC parser generator builds up

SLR parsing table
Canonical LR parsing table
LALR parsing table
None of the given

Question No: 1 (Marks: 1) - Please choose one
LR(1) parsing is --- base parsing.

DFA
CFG
PDA
None of the given

Question No: 1 (Marks: 1) - Please choose one
The LR(1) parsers can not recognize precisely those languages in which one-symbol lookahead suffices to determine whether to shift or reduce.

True
False

Question No: 1 (Marks: 1) - Please choose one
Yacc contains built-in support for handling ambiguous grammars resulting in shift-reduce conflicts. By default these conflicts are solved by performing the _____.

Shift action
Reduce action
Shift and reduce actions
De-allocation of memory

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Question No: 1 (Marks: 1) - Please choose one
S --> A | xb A --> aAb | x This grammar contains a reduce-reduce conflict.

True
False

Question No: 1 (Marks: 1) - Please choose one
S --> a | B
B --> Bb | E The non-terminal _____ is left recursive.

B
a
E
None of the given

Question No: 1 (Marks: 1) - Please choose one
Following statement represents: if x relop y goto L

abstract jump
Conditional jump
While loop
None of the Given

Question No: 1 (Marks: 1) - Please choose one
When generating a lexical analyzer from a _____ description, the item sets (states) are constructed by two types of "moves": character moves and e moves.

Character
Grammar
Token
Sentence

Question No: 1 (Marks: 1) - Please choose one
Left factoring is enough to make a grammar LL(1).

True
False

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Question No: 1 (Marks: 1) - Please choose one
Register allocation by graph coloring uses a register interference graph. _____ nodes in the graph are joined by an edge when the live ranges of the values they represent overlap.

Two p116
Three
Four
Five

Question No: 1 (Marks: 1) - Please choose one
S --> A B A --> e | aA B --> e | bB - FIRST(S) contains ___ elements.

3
4
5
6

Question No: 1 (Marks: 1) - Please choose one
The notation _____ instructs YACC to push a computed attribute value on the stack.

$$ Page no : 98
&&
##
--

Question No: 1 (Marks: 1) - Please choose one
Simple code generation considers one AST node at a time. If the target is a register machine, the code can be generated in one _____ traversal of the AST, possibly introducing temporaries when running out of registers.

Depth-first
Breadth-first
Top-Down
Bottom-Up

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Question No: 1 (Marks: 1) - Please choose one
Grammars with LL(1) conflicts can be made LL(1) by applying left-factoring, substitution, and left-recursion removal. Left-factoring takes care of _____conflicts.

FIRST/FIRST
FIRST/SECOND
SECOND/FIRST
None of the given

Question No: 1 (Marks: 1) - Please choose one
In an attribute grammar each production rule(N--> a) has a corresponding attribute evaluation rule that describes how to compute the values of the _____attributes of each particular node N in the AST.

Synthesized page no : 92
Complete
Free
Bounded

Question No: 1 (Marks: 1) - Please choose one
When constructing an LR(1) parser we record for each item exactly in which context it appears, which resolves many conflicts present in _____parsers based on FOLLOW sets.

SLR(1)
LRS(1)
RLS(1)
None of the given

Question No: 1 (Marks: 1) - Please choose one
The _____translation statements can be conveniently specified in YACC

Syntax-directed Page no : 120
Image-directed
Sign-directed
None of the given.

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Question No: 1 (Marks: 1) - Please choose one
Backpatching to translate flow-of-control statements in ____ pass.

one
two
three
all of the given

Question No: 1 (Marks: 1) - Please choose one
Alternative of the backtrack in parser is Look ahead symbol in _____ .

Input
Output
Input and Output
None of the given

Question No: 1 (Marks: 1) - Please choose one
Typical compilation means programs written in high-level languages to low-level _____.

Object code
Byted code
Unicode
Both Object Code and byte code

Question No: 1 (Marks: 1) - Please choose one
In PASCAL _____ represent the inequality test.

:
:=
=
<>
None of the given

Question No: 1 (Marks: 1) - Please choose one
LR parsing _____ a string to the start symbol by inverting productions.

Reduces
Shifts
Adds
None of the given

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Question No: 1 (Marks: 1) - Please choose one
In multi pass compiler during the first pass it gathers information about _____ .

Declaration
Bindings
Static information
None of the given

Question No: 1 (Marks: 1) - Please choose one
_____ phase which supports macro substitution and conditional compilation.

Semantic
Syntax
Preprocessing
None of given

Question No: 1 (Marks: 1) - Please choose one
In parser the two LL stand(s) for _____ .

Left-to-right scan of input
left-most derivation
All of the given
None of the given

Question No: 1 (Marks: 1) - Please choose one
Parser always gives a tree like structure as output

True
False

Question No: 1 (Marks: 1) - Please choose one
Lexer and scanner are two different phases of compiler

True
False

Question No: 1 (Marks: 1) - Please choose one
_____tree in which each node represents an operator and children of the node represent the operands.

Abstract syntax Page no : 100
Concrete syntax

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Parse
None of the given

Question No: 1 (Marks: 1) - Please choose one
In compilation process Hierarchical analysis is also called

Parsing
Syntax analysis
Both Parsing and Syntax analysis
None of given

Question No: 1 (Marks: 1) - Please choose one
Ambiguity can easily be handled by Top-down Parser
Select correct option:

True
False

Question No: 1 (Marks: 1) - Please choose one
Front-end of a two pass compiler is consists of Scanner.

True
False

Question No: 1 (Marks: 1) - Please choose one
LL(1) parsing is called non-predictive parsing.

True
False

Question No: 1 (Marks: 1) - Please choose one
In predictive parsing table the rows are _____ .

Non-terminals
Terminals
Both non-terminal and terminals
None of the given

Question No: 1 (Marks: 1) - Please choose one
In LL1() parsing algorithm _____ contains a sequence of grammar symbols.

Stack
Link list

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Array
None

Question No: 1 (Marks: 1) - Please choose one
Consider the grammar
A --> B C D
B --> h B | epsilon
C --> C g | g | C h | i
D --> A B | epsilon
First of C is _____ .
Select correct option:

g, I look down for reference
g
h i
i

Question No: 1 (Marks: 1) - Please choose one
AST summarizes the grammatical structure with the details of derivations.

True
False

Question No: 1 (Marks: 1) - Please choose one
Left factoring is enough to make LL1 grammar

True
False

Question No: 1 (Marks: 1) - Please choose one
f X is a non-terminal in A? aX•?, then the interpretation of this transition is more complex because non-terminals do not appear in input

Yes
No

Question No: 1 (Marks: 1) - Please choose one
If / is a set of items for a grammar then closure (/) is a set of items constructed from / by the following rule.

If A --> aX.Y is in closure (/) and Y --> r is production, then add X --> .r to closure (/).
If A --> a.XY is in closure (/) and X --> r is production, then add X --> .r to closure (/).
If A --> aXY. is in closure (/) and A --> r is production, then add X --> .r to closure (/).

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None of these

Question No: 1 (Marks: 1) - Please choose one
NFA of LR(0) items means _____

look-ahead one sybole
no look-ahead
look-ahead all sybols
None of the given

Question No: 1 (Marks: 1) - Please choose one
A grammar is LR if a ------ shift reduce-reduce parser can recognize handles when they appear on the top of stack.

left-to-reverse
left-to-rise
left-to-right
None of the given.

Question No: 1 (Marks: 1) - Please choose one
The output from the algorithm of constructing the collection of canonical sets of LR(1) items will be the _____

Original Grammar G
Augmented grammar G'
Parsing table
None of the given

Question No: 1 (Marks: 1) - Please choose one
Reduction of a handle to the ------- on the left hand side of the grammar rule is a step along the reverse of a right most derivation.

Terminal
Non-terminal

Question No: 1 (Marks: 1) - Please choose one
NFA of LR(1) items means _____

no look-ahead
look-ahead one sybole

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look-ahead all sybols
None of the given

Question No: 1 (Marks: 1) - Please choose one
In canonical collection procedure a DFA can not be constructed from NFA using the subset construction, similar to one we used for lexical analysis.

True
False

Question No: 1 (Marks: 1) - Please choose one
performing common subexpression elimination on aa dependncy graph requires the identification of nodes with the same operator and operands.when using a hash table (with a hash function based on operator and operands) all_____ nodes can be identified in linear time.

common
uncommon
next
previous

Question No: 1 (Marks: 1) - Please choose one
Linear IRs resemble pseudo-code for same _____.

Automated Machine
Mechanical machines
Token machines
Abstract machine

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Question No: 1 (Marks: 1) - Please choose one
The regular expressions a*|b* and (a|b)* describe the _____set of strings.

Same
Different
Onto

Question No: 1 (Marks: 1) - Please choose one
Back patching to translate flow-of-control statements in ____ pass.

one Page no : 111
two
three
all of the given

Question No: 1 (Marks: 1) - Please choose one
Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string "abbcde" can be parsed bottom-up by the following reduction steps: (i) aTbcde (ii) aTde (iii) aTUe (iv) S So what can be a handle from the following?

The second (b) in (abbcde)
The first (b) in (abbcde)
The substring (cd) in (abbcde)
None of the given

Question No: 1 (Marks: 1) - Please choose one
Yacc contains built-in support for handling ambiguous grammars resulting in _____ conflicts.

Shift-reduce
Shift-Shift
Shift-second
None of the given

Question No: 1 (Marks: 1) - Please choose one
A lexical analyzer generator automatically constructs a _____ that recognizes tokens.
:

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FA
PDA
DP
None of the given

Question No: 1 (Marks: 1) - Please choose one
Attributes whose values are defined in terms of a node's own attributes, node's siblings and node's parent are called _____ .

Inherited attributes Page no : 92
Physical attributes
Logical attributes
Un-synthesized attributes

Question No: 1 (Marks: 1) - Please choose one
The following two items A -> P • Q B -> P • Q can co-exist in an _____ item set.

LR
LS
LT
PR

Question No: 1 (Marks: 1) - Please choose one
Three-address codes are often implemented as a _____.

Set of quadruples Page no : 104
Set of doubles
Set of Singles
None of the given

Question No: 1 (Marks: 1) - Please choose one
The error handling mechanism of the yacc parser generator pushes the input stream back when inserting 'missing' tokens.

True
False

Question No: 1 (Marks: 1) - Please choose one
Flow of values used to calculate synthesized attributes in the parse tree is:

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Bottom-up Page no: 92
Right to left
Top-Down
Left to right

Question No: 1 (Marks: 1) - Please choose one
What does following statement represent? x[i] = y

Prefix assignment
Postfix assignment
indexed assignment Page no : 107
None of the given

Question No: 1 (Marks: 1) - Please choose one
A lexical analyzer transforms a stream of tokens. The tokens are stored into symbol table for further processing by the parser.

True Page no: 99
False

Question No: 1 (Marks: 1) - Please choose one
LR parsers can handle _____ grammars.

Left-recursive Page no: 163
file-recursive
End-recursive
Start-recursive

Question No: 1 (Marks: 1) - Please choose one
_____ convert the reloadable machine code into absolute machine code by linking library and reloadable object files.

Assembler
Loader/link-editor
Compiler
Preprocessor

Question No: 1 (Marks: 1) - Please choose one
Consider the following grammar,

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A --> B C D
B --> h B | episilon
C --> C g | g | C h | i
D --> A B | episilon

First of A is _____ .
h, g, i
gh
None of the given

Question No: 1 (Marks: 1) - Please choose one
One of the core tasks of compiler is to generate fast and compact executable code.

True
False

Question No: 1 (Marks: 1) - Please choose one
Compilers are sometimes classified as.

Single pass
Multi pass
Load and go
All of the given

Question No: 1 (Marks: 1) - Please choose one
In multi pass compiler during the first pass it gathers information about _____ .

Declaration
Bindings
Static information
None of the given **

Question No: 1 (Marks: 1) - Please choose one
For each language to make LL(1) grammar, we take two steps, 1st is removing left recurrence and 2nd is applying fin sequence.

True
False

Question No: 1 (Marks: 1) - Please choose one
_____is evaluated to yield a value.

Command

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Expression
Declaration
Declaration and Command

Question No: 1 (Marks: 1) - Please choose one
We can get an LL(1) grammar by _____ .
Removing left recurrence
Applying left factoring
Removing left recurrence and Applying left factoring
None of the given

Question No: 1 (Marks: 1) - Please choose one
Can a DFA simulate NFA?

Yes
No
Sometimes
Depend upon nfa

Question No: 1 (Marks: 1) - Please choose one
Which of the statement is true about Regular Languages?

Regular Languages are the most popular for specifying tokens.
Regular Languages are based on simple and useful theory.
Regular Languages are easy to understand.
All of the given

Question No: 1 (Marks: 1) - Please choose one
The transition graph for an NFA that recognizes the language (a | b)*abb will have following set of states.

{0}
{0,1}
{0,1,2}
{0,1,2,3} not sure

Question No: 1 (Marks: 1) - Please choose one
Functions of Lexical analyzer are?

Removing white space
Removing constants, identifiers and keywords
Removing comments
All of the given

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Question No: 1 (Marks: 1) - Please choose one
Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string "abbcde" can be parsed bottom-up by the following reduction steps: (i) aTbcde (ii) aTde (iii) aTUe (iv) S So, what can be a handle from the following?

The whole string, (aTUe) Page no : 68
The whole string, (aTbcde)
The whole string, (aTde)
None of the given

Question No: 1 (Marks: 1) - Please choose one
The LR(1) items are used as the states of a finite automaton (FA) that maintains information about the parsing stack and progress of a shift-reduce parser.

True Page no: 74
False

Question No: 1 (Marks: 1) - Please choose one
Flex is an automated tool that is used to get the minimized DFA (scanner).

True
False Page no: 26

Question No: 1 (Marks: 1) - Please choose one
We use ----- to mark the bottom of the stack and also the right end of the input when considering the Stack implementation of Shift-Reduce Parsing.

Epsilon
#
$ Page no : 65
None of the given

Question No: 1 (Marks: 1) - Please choose one
When generating a lexical analyzer from a token description, the item sets (states) are constructed by two types of "moves": character moves and ____ moves.

E (empty string) Page no : 18
#
@

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none of given

Question No: 1 (Marks: 1) - Please choose one
Bottom-up parsers handle a _____ class of grammars.

large Page no : 63
small
medium
none of the given

Question No: 1 (Marks: 1) - Please choose one
Let a grammar G = (Vn, Vt, P, S) is modified by adding a unit production S'--> S to the grammar and now starting non-terminals becomes S' and grammar becomes G' = (Vn U {S'}, Vt, PU{S' --> S}, S'). The Grammar G' is called the ----------

Augmented Grammar Page no : 76
Lesser Grammar
Anonymous Grammar
none of given

Question No: 1 (Marks: 1) - Please choose one
Parser takes tokens from scanner and tries to generate _____ .

Binary Search tree
Parse tree
Syntax trace Page no : 6
None of the given

Question No: 1 (Marks: 1) - Please choose one
In Flex specification file different sections are separated by _____ .

%% Page no: 26
&&
##
\\

Question No: 1 (Marks: 1) - Please choose one
Consider the grammar A --> B C D

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B --> h B | epsilon
C --> C g | g | C h | i
D --> A B | epsilon

Follow of B is _____ .

h
g, h, i, $
g, i
g

Question No: 1 (Marks: 1) - Please choose one
Consider the grammar A --> B C D
B --> h B | epsilon
C --> C g | g | C h | i
D --> A B | epsilon

Follow of C is _____ .

g, h, i, $ Page no : 47
g, h, $
h, i, $
h, g, $

Question No: 1 (Marks: 1) - Please choose one
In DFA minimization we construct one _____ for each group of states from the initial DFA.

State Page no : 25
NFA
PDA
None of given

Question No: 1 (Marks: 1) - Please choose one
An important component of semantic analysis is _____ .

code checking
type checking page no : 6
flush checking
None of the given

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Question No: 1 (Marks: 1) - Please choose one
Intermediate Representation (IR) stores the value of its operand in _____ .

Registers Page no : 10
Memory
Hard disk
Secondary storage

Question No: 1 (Marks: 1) - Please choose one
In _____certain checks are performed to ensure that components of a program fit together meaningfully.
Linear analysis
Hierarchical analysis
Semantic analysis Page no : 33
None of given

Question No: 1 (Marks: 1) - Please choose one
Which of the following statement is true about Two pass compiler.

Front End depends upon Back End
Back End depends upon Frond End page no : 5
Both are independent of each other
None of the given

Question No: 1 (Marks: 1) - Please choose one
_____ algorithm is used in DFA minimization.

James's
Robert's
Hopcroft's Page no : 19
None of given

Question No: 1 (Marks: 1) - Please choose one
A _____ is a top down parser.
Predictive Parsing Page no: 46
Reactive parser
Proactive parser
None of the given

Question No: 1 (Marks: 1) - Please choose one
Lexical Analyzer generator _____ is written in Java.

Flex
Jlex Page no : 26

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Complex
None of given

Question No: 1 (Marks: 1) - Please choose one
_____avoid hardware stalls and interlocks.

Register allocation
Instruction scheduling Page no : 10
Instruction selection
None of given

Question No: 1 (Marks: 1) - Please choose one
Recursive _____ parsing is done for LL(1) grammar.

Decent Page no : 47
Ascent
Forward
Backward

Question No: 1 (Marks: 1) - Please choose one
Left factoring of a grammar is done to save the parser from back tracking.

True Page no:61
False

Question No: 1 (Marks: 1) - Please choose one
Responsibility of _____ is to produce fast and compact code.

Instruction selection
Register allocation
Instruction scheduling
None of given Page no: 9

Question No: 1 (Marks: 1) - Please choose one
Optimal registers allocation is an NP-hard problem.

True
False Page no : 10

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Question No: 1 (Marks: 1) - Please choose one
Front end of two pass compiler takes_____ as input.

Source code Page no: 5
Intermediate Representation (IR)
Machine Code
None of the Given

Question No: 1 (Marks: 1) - Please choose one
In Three-pass compiler _____ is used for code improvement or optimization.

Front End
Middle End Page no : 10
Back End
Both Front end and Back end

Question No: 1 (Marks: 1) - Please choose one
_____ of a two-pass compiler is consists of Instruction selection, Register allocation and Instruction scheduling.

Back end Page no : 9
Front end
Start
None of given

Question No: 1 (Marks: 1) - Please choose one
NFA is easy to implement as compared to DFA.

True
False Page no : 19

Question No: 1 (Marks: 1) - Please choose one
In Back End module of compiler, optimal register allocation uses_____ .

O(log n)
O(n log n)
N P-Complete Page no : 10
None of the given

Question No: 1 (Marks: 1) - Please choose one
In a transition table cells of the table contain the _____ state.

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Reject state
Next state Page no 18
Previous state
None of the given

Question No: 1 (Marks: 1) - Please choose one
Parser generator for the grammar LALR (1) is:
YACC, Bison, CUP Page no: 88

Question No: 1 (Marks: 1) - Please choose one
Attributes of a node whose values are defined wholly in terms of attributes of node's children and from constants are called _____.

Synthesized attributes Page no : 92

Question No: 1 (Marks: 1) - Please choose one
Goto L statement represent

Unconditional jump Page no : 107

Question No: 1 (Marks: 1) - Please choose one
Dotted items (T□a •b) record which part of a token has already been matched. Integer? ([0-9])+ • This is a _____ item.

Extended Page no : 73

Question No: 1 (Marks: 1) - Please choose one
If T --> XYZ is a production of grammar G then which of the following item indicates that a string derivable from X has been seen so far on the input and we hope to see a string derivable from YZ next on the input.

Question No: 1 (Marks: 1) - Please choose one
The most powerful parser is:

Question No: 1 (Marks: 1) - Please choose one
In the Parsing Table the rows correspond to Parsing DFA states and columns correspond to ----.

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1. _____convert the relocatable machine code into absolute machine codeby linking library and relocatable objectfiles.

   - Assembler
   - Loader/link-editor
   - Compiler
   - Preprocessor

2. Parsers take__as input from lexical analyzer.
   - Linker
   - Token
   - Instruction
   - None of the given

3. The regular expression_denotes, the set of all strings of a's and b's of length two
   - a*
   - (a*|b*)*
   - (a*b*)*
   - (a|b)(a|b)

4. _____is a regular expression for the set of all strings over the alphabet {a} that has an even number of a's.
   - aa*
   - (aa)*
   - aa*a
   - a(aa)*

5. _____Phase supports macro substitution and conditional compilation.
   - Semantic
   - Syntax
   - Preprocessing
   - None of given

6. In LL(1) parsing algorithm,_contains a sequence of grammar symbols.
   - Stack PG # 62
   - Link List
   - Array
   - None of the given.

7. Consider the grammar

A → B C D
B → h B | €
C → C g | g | C h | i
D→ AB | £

First of A is_____.

- h, g, i
- g
- h
- None of the given.

8. _____parsers never shifts into an error state.
   - LS
   - LT
   - LR
   - LP
9. In parser, the two LL stand for___.
   - Left-to-right scan of input
   - left-most derivation
   - Left-to-right scan of input and left-most derivation PG # 54
   - None of the given
10. is elaborated to produce bindings.
    - Declaration
    - Expression
    - Command
    - None of the given
11. A lexical analyzer generated by___is essentially a FSA.
    - Dex
    - Mex
    - Fex
    - Lex
12. A lexical analyzer generated by lex is essentially a PDA (Push Down Automaton).
    - True
    - False
13. T

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he actions (shift, reduce) in a SLR(1) parser depend on a look ahead symbol (_____)

- Current input token
- Next Input Token
- Previous output Token
- Previous Input Token.

14. The following grammar contains a conflict. S ⟶ *A* | *xb*

- Shift-Reduce
- First-Reduce
- Shift-First
- Reduce-Reduce

15. S → A | xb
A → aAb | x
This grammar contains a_____conflict.

- Shift-Reduce
- First-Reduce
- Shift-First
- Reduce-Reduce

16. Consider the Following
S ⟶ AB

- 1
- 2
- 3
- 4

17. ___is a register allocation technique that *always* finds the minimal number of registers needed for aprocedure.

- Dangling reference
- Graph coloring
- Left Factoring
- Right Recursion

18. Graph coloring is a register allocation technique that operates at *individual* basic blocks.

- True
- False

19. Graph coloring is a register allocation heuristic that *usually* finds he minimal number of registers needed for aprocedure.

- True
- False

20. S → a S | Sa | c
This grammar is ambiguous.

- True
- False

21. When generating code at the basic block level, the dependency graph must be converted to target code. By identifying , instruction selection and instruction ordering can be performed efficiently in a single pass.

- Ladder sequences
- Physical sequences
- Logical sequences
- Token sequences

22. _can be considered a small compiler since it transforms a source language (assembly) into a lessabstract target language (binary object code)

- Parser
- Assembler
- Lexical analyzer
- Scanner

23. When memory allocator operates on chunks which include some administrative part and a block of user data. Theadministrative part includes____flag for marking the chunk as free or in-use.

- One
- Two
- Three
- Four

24. parser transforms a stream of tokens into an_____.

- AST

- IST
- EST
- ATS

25. The parser generator yacc can handle_____grammars
    - LL(1)
    - LT(1)
    - LS(1)
    - LF(1)

26. The parser generator yacc can handle LL(1) grammars.
    - True
    - False

27. The yacc parser generator can handle LALR(1) grammars.
    - True
    - False

28. Simple code generation considers one AST node at a time. If the target is a *register* machine, the code can be generated in one__traversal of the AST, possibly introducing temporaries when running out of registers.
    - Depth-first
    - Breadth-first
    - Depth-second
    - Breadth-second

29. A linker combines multiple object files into a_____executable object.
    - Single
    - Double
    - Triple
    - Quadruple

30. The notation____instructs YACC to push a computed attribute value on the stack.
    - $$ PG # 106
    - &&
    - ##
    - --

31. The following two items

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A → P • Q
B → P • Q
can co-exist in an___item set

- LR
- LS
- LT
- PR

32. When generating a lexical analyzer from a_____description, the item sets (states) are constructed by twotypes of "moves": character moves and € moves.

- Character
- Grammar
- Token
- Sentence

33. Hybrid IRs combine elements of_____

- Graphical (structural)
- Linear IRs
- Both graphical and linear IRs PG # 108
- Non-Linear IRs

34. x[i] = y This is___.

- Prefix assignment
- Postfix assignment
- Index assignment PG # 115
- Non-Index assignment

35. A lexical analyzer generator automatically constructs a_____that recognizes tokens.

- FA PG # 18
- PDA
- DP
- Unidirectional Graph

36. if x relop y goto L Above statement is _____

- Abstract jump

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- Conditional jump PG # 115
- While loop
- Unconditional jump

37. I
n a CFG (Context Free Grammar) the set of terminal and non-terminal symbols must be.

- Disjoint
- Logical
- Relational
- Joint

38. S → a | B
B → Bb | €
The non-terminal B is left recursive.

- True
- False

39.
YACC contains built-in support for handling ambiguous grammars resulting in_conflicts.

- Shift-reduce
- Shift-Shift
- Reduce-reduce
- Reduce-Shift
- Segment-directed

43. When constructing an LR(1) parser we record for each item exactly in which context it appears, which resolvesmany conflicts present in_ parsers based on FOLLOW sets.

- SLR(1)
- LRS(1)
- RLS(1)
- SLL(1)

44. Code generation module has to tackle_____.

- Memory management
- Instruction selection
- Instruction scheduling
- All of the given PG # 129

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45. For convenience, lexical analyzers should read the complete program into memory.

- Input
- Output
- Input and output
- Tokens

40. Considering the following grammar:

S → A | x
A → aAb | x
The grammar contains a___conflict.

- Reduce-reduce
- First-first
- Shift-shift
- Shift-reduce

41. SLR (1) parsers only reduce a production rule when the current input token is an element of the FOLLOW set ofthat rule.

S → A B

A → € | aA

A → b | bB

- FOLLOW (A) contains 2 elements.

- True
- False

42. SLR (1) parsers only reduce a production rule when the current input token is an element of the FOLLOW set ofthat rule.

S → A B
A → a | aA
B→€|bB

- FOLLOW (A) contains 2 elements.

- True
- False

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43. The order in which the DAG is traversed can lead to_____code

- Better PG # 143
- Worse
- Large
- Garbage

44. Register allocation problem uses the strategy of_____.

- Graph coloring PG # 144
- Graph nodding
- Graph edging
- Graph patching

48. Typical compilation means programs written in high-level languages to low-level

- Object code PG # 06
- Byted code
- Unicode
- Object code and byte code

45. In compilation process, Hierarchical analysis is also called_.

- Parsing
- Syntax analysis.
- Parsing and syntax analysis
- None of the given

46. IR (Intermediate Representation) stores the value of its operand in____.

- Registers PG # 10
- Memory
- Hard disk
- None of the given

47. exeme is a sequence of characters in the source program that is matched by the pattern for a.

- Linker

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- Token
- Control flow
- None of the given

48. Parsers take___ as input from lexical analyzer.

- Linker
- Token
- Instruction
- None of the given

49. What kind of abstract machine can recognize strings in a regular set?

- DFA
- NFA
- PDA
- None of the given

53. In DFA minimization, we construct one__for each group of states from the initial DFA.

- State PG # 30
- NFA
- PDA
- None of the given

50. _____(Lexical Analyzer generator), is written in java.

- Flex
- Jlex PG # 31
- Complex
- None of the given

51. In Flex specification file, different sections are separated by_____.

- %% PG # 31
- &&
- ##
- None of the given

52. Recursive__parsing is done for LL(1) grammar.

- Decent
- Ascent

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- Forward
- None of the given

56. Alternative of the backtrack in parser is Look ahead symbol in_.

- Input
- Output
- Input and output
- None of the given

53. Parser takes tokens from scanner and tries to generate_____

- Binary search tree
- Parse tree
- Binary search tree and parse tree.
- None of the given

54. In predictive parsing table, the rows represents_____.

- Terminals
- Both non-terminal and terminal
- Non-terminal PG # 62
- None of the given

55. A predictive parser is a top-down parser.

- True
- False

56. In LL(1) parsing algorithm, contains a sequence of grammar symbols.

- Stack PG # 62
- Link list
- Array
- None of the given

57. Bottom-up parsing uses only___kinds of actions.

- Two PG # 71
- Three
- Four
- Five

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58. Bottom-up parsers handle a_class grammar.

- Large PG # 49
- Small
- Medium
- None of the given

59. The shift action a terminal on the stack.

- Pushes PG # 73
- Pops
- Both push and pops
- None of the given

60. Reduce action zero or more symbols from the stack.

- Pushes
- Pops PG # 73
- Both push and pops
- None of the given

61. In compilers, linear analysis is also called_____.

- Lexical analysis
- Scanning
- Lexical analysis and scanning
- None of the given

62. Back End of two-pass compiler uses_____algorithm.

- O(n)
- O(n log n)
- NP Complete
- None of the given

63. The Back End of a compiler consist of_____.

- Instruction selection
- Register allocation

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- Instruction scheduling
- All of the given

64. In Back End module of compiler, optimal register allocation uses_.
- O(log n)
- O(n log n)
- NP-Complete
- None of the given

65. lexeme is a sequence of characters in the source program that is matched by the pattern for a.
- Linker
- Token
- Control flow
- None of the given

66. _____is a regular expression for the set of all strings over the alphabets {a} that has an even number of a's.
- aa*
- (aa)*
- aa*a
- a(aa)*

67. ____algorithm is used in DFA minimization.
- James's
- Robert's
- Hopcroft's PG # 25
- None of the given

68. _is an important component of semantic analysis.
- Code checking
- Type checking PG # 39
- Flush checking
- None of the given

69. In , certain checks are performed to ensure that components of a program fit together meaningfully.
- Linear analysis
- Hierarchical analysis
- Semantic analysis

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- None of the given

70. _____read the input character and produce sequence of tokens as output.
    - Lexical analyzer
    - Parser
    - Symbol table
    - None of the given

71. _____of a two-pass compiler is consist of instruction selection, Register allocation and instructionscheduling.
    - Backend
    - Frontend
    - Start
    - None of the given

72. _____is evaluated to yield a value.
    - Command
    - Expression
    - Declaration
    - None of the given

73. A parser transforms a stream of tokens into an AST (Abstract Syntax Tree).
    - True
    - false

74. A parser transforms a stream of characters into a stream of tokens.
    - True
    - False

75. A lexical analyzer transforms a stream of characters into a stream of tokens.
    - True
    - False

76. S → a | A
    A → Aa | a
    This grammar is ambiguous.
    - True
    - False

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77. The regular expressions (a+|b)? and a+|b? describe the same set of strings.
    - True
    - False
78. The regular expressions a*|b* and (a|b)* describe the same set of strings.
    - True
    - False
79. The regular expressions a+a and a*aa describe the same set of strings.
    - True
    - False
80. A lexical analyzer *generator* automatically construct a FSA (Finite State Automaton) that recognizes tokens.The generator is driven by a regular description
    - True
    - False
81. The transition table in a lexical analyzer records for each state (row) which token, if any, is recognized in thatstate. - For each token there may be more than one "recognizing" row in the table.
    - True
    - False
82. A recursive descent parser is based on a PDA (Push Down Automaton).
    - True
    - False
83. A bottom-up parser creates the nodes in the AST in pre-order.
    - True
    - False
84. A top-down parser creates the nodes in the AST (Abstract Syntax Tree) in preorder.
    - True
    - False
85. A_____parser creates the nodes in the AST in preorder.
    - Top – Down
    - Bottom – Up

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- Middle – Ware
- Straight

86. The stack used in a bottom-up parser contains an alternating sequence of states and grammar symbols.
    - True
    - False

87. The following two items
    A→ P• Q
    A→ PQ •
    Can coexist in an LR item set.
    - True
    - False

88. The Following two Items
    A → x • B
    B → • y
    Can coexist in an LR item set.
    - True
    - False

89. The Following two Items
    B → P • P
    B → Q • Q
    Can coexist in an LR item set.
    - True
    - False

90. S → A | xb
    A → aAb | x
    This is an LALR(1) grammar.
    - True
    - False

91. A linker combines multiple object files into a single executable object.
    - True
    - False

92. Data-flow equations can be solved efficiently by using bitwise boolean instructions (AND, OR, etc.).

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- True
- False

93. Data-flow equations operate with IN, OUT, GEN, and KILL sets.
    - True
    - False

94. When threading an AST it might be necessary to introduce additional (join) nodes to ensure that each languageconstruct has a single exit point.
    - True
    - False

95. An iterative interpreter operates on a threaded AST.
    - True
    - False

96. S → A | B
    A → € | aA
    B → b | bB
    FIRST(S) contains_____elements.
    - 2
    - 3
    - 4
    - None

97. The following set
    S → • A x {$}
    A → • a {x}
    A → • aA {x}
    is a valid LR(1) item set
    - True
    - False

98. S → Ab
    A → Aa | €
    - True
    - False

99. The regular expressions a(b|c) and ab|ac describe the same set of strings.
    - True

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- False

100. S → a | B
B → Bb | E
The non-terminal___ is left recursive.

- B
- a
- E
- None of the given

101. In PASCAL_____ represent the inequality test.

- :=
- =
- <>
- None of the given

102. In parser the two LL stand(s) for_____.

- Left-to-right scan of input
- left-most derivation
- All of the given
- None of the given

103. Consider the grammar
A →B C D
B → h B | epsilon
C → C g | g | C h | i
D → AB | epsilon
First of C is_____.

- g, I
- g
- h , i
- i

104. Three-address codes are often implemented as a __.

- Set of quadruples PG # 104
- Set of doubles
- Set of Singles
- None of the given

105. What does following statement represent? x[i] = y

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- Prefix assignment
- Postfix assignment
- indexed assignment PG #107
- None of the given

106. _____convert the reloadable machine code into absolute machine code by linking library and reloadable objectfiles.
- Assembler
- Loader/link-editor
- Compiler
- Preprocessor

107. Consider the following grammar,

A → B C D
B → h B | epsilon
C → C g | g | C h | i
D → AB | epsilon

First of A is_____.
- h, g, i
- g
- h
- None of the given

108. One of the core tasks of compiler is to generate fast and compact executable code.
- True PG # 14
- False

109. Compilers are sometimes classified as.
- Single pass
- Multi pass
- Load and go
- All of the given

110. In multi pass compiler during the first pass it gathers information about
- Declaration

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- Bindings
- Static information
- None of the given

111. We can get an LL(1) grammar by_____.

- Removing left recurrence
- Applying left factoring
- Removing left recurrence and Applying left factoring
- None of the given

112. Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string "abbcde" can be parsedbottom-up by the following reduction steps:

(i) aTbcde
(ii) aTde
(iii) aTUe
(iv) S

So, what can be a handle from the following?

- The whole string, (aTUe) PG # 68
- The whole string, (aTbcde)
- The whole string, (aTde)
- None of the given

113. When generating a lexical analyzer from a token description, the item sets (states) are constructed by two typesof "moves": character moves and_____moves.

- E (empty string) PG # 18
- #
- @
- none of given

114. Which of the following statement is true about Two pass compiler.

- Front End depends upon Back End

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- Back End depends upon Frond End PG # 5
- Both are independent of each other
- None of the given

115. avoid hardware stalls and interlocks.
- Register allocation
- Instruction scheduling PG #10
- Instruction selection
- None of given

116. Front end of two pass compiler takes_____as input.
- Source code PG # 5
- Intermediate Representation (IR)
- Machine Code
- None of the Given

117. In Three-pass compiler _____is used for code improvement or optimization.
- Front End
- Middle End PG # 10
- Back End
- Both Front end and Back end

118. _____of a two-pass compiler is consists of Instruction selection, Register allocation andInstruction scheduling.
- Back end PG # 9
- Front end
- Start
- None of given

119. NFA is easy to implement as compared to DFA.
- True
- False PG # 19

120. In a transition table cells of the table contain the___state.

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- Reject state
- Next state PG #18
- Previous state
- None of the given

121. The regular expressions a*|b* and (a|b)* describe the____set of strings.
- Same
- Different
- Onto

122. A canonical collection of sets of items for an augmented grammar, C is constructed as

- For each set / in C and each grammar symbol X where goto (C, X) is empty and not in C add the set goto (C, X)to C.
- The first set in C is the closure of {[S' --> .S]}, where S' is starting symbol of original grammar and S isthe starting non-terminal of augmented grammar. PG # 72
- The first set in C is the closure of {[S' --> .S]}, where S is starting symbol of original grammar and S' is theStarting non-terminal of original grammar.

123. The_____translation statements can be conveniently specified in YACC
- Syntax-directed PG # 120
- Image-directed
- Sign-directed
- None of the given.

124. Attributes whose values are defined in terms of a node's own attributes, node's siblings and node's parent arecalled_.
- Inherited attributes PG # 92
- Physical attributes
- Logical attributes

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- Un-synthesized attributes

125. Consider the grammar

A → B C D
B → h B | epsilon
C → C g | g | C h | i
D → AB | epsilon

Follow of B is_____.

- h
- g, h, i, $
- g, i
- g

126. Consider the grammar A --> B C D

A → B C D
B → h B | epsilon
C → C g | g | C h | i
D → AB | epsilon

Follow of C is_____.

- g, h, i, $ PG # 47
- g, h, $
- h, i, $
- h, g, $

127. The test of string is described by a rule called a, associated with token.

- Character
- Loader
- Pattern
- None of the given

128. Bottom up parsing is also called____.

- LR Parsing PG # 70

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- LT Parsing
- LS Parsing
- None of the given

129. A DFA can be reconstructed from NFA using the subset construction, similar to one used for
- Lexical Analysis PG # 82
- Physical Analysis
- Logical Analysis
- Parsing

130. Which of the following system software resides in the main memory always?
- Text editor
- Assembler
- Linker
- Loader

131. plays an important role in code optimization.
- DAG PG # 143
- Lexical Analyzer
- AGD
- Memory Management

132. LR parsers can handle_____grammars.
- Left-recursive PG # 63
- file-recursive
- End-recursive
- Start-recursive

133. Performing common sub expression elimination on a dependency graph requires the identification of nodeswith the same operator and operands. When using a hash table (with a hash

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function based on operator and operands) all___nodes can be identified in linear time.

- Common
- Uncommon
- Next
- Previous

134. Linear IRs resembles pseudo-code for same__.

- Automated Machine
- Mechanical machines
- Token machines
- Abstract machine PG # 100

135. Responsibility of _____is to produce fast and compact code.

- Instruction selection
- Register allocation
- Instruction scheduling
- None of given

136. Optimal registers allocation is an NP-hard problem.

- True
- False Page no : 10

137. Left factoring of a grammar is done to save the parser from back tracking.

- True Page no:61
- False

138. Recursive _____parsing is done for LL(1) grammar.

- Decent Page no : 47
- Ascent
- Forward
- Backward

139. If X is a terminal in A--> aX•?, then this transition corresponds to a shift of_____from input to top of parse stack.

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- X
- A
- a
- None of the given

140. An ----- does not need to examine the entire stack for a handle, the state symbol on the top of the stack contains all the information it needs.

- LR parser
- RL parser
- BU parser
- None of the given

141. Suppose ? begins with symbol X which may be a terminal (token) or non-terminal. The item can be written as A? Xa•?.

- True
- False

142. YACC parser generator builds up

- SLR parsing table
- Canonical LR parsing table
- LALR parsing table
- None of the given

143. LR(1) parsing is--- base parsing.

- DFA
- CFG
- PDA
- None of the given

144. The LR(1) parsers can not recognize precisely those languages in which one-symbol lookahead suffices to determine whether to shift or reduce.

- True
- False

145. S --> A | xb A --> aAb | x This grammar contains a reduce-reduce conflict.

- True

False

146. Following statement represents: if x relop y goto L

- abstract jump
- Conditional Jump

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- While Loop
- None Of Given

147. Left factoring is enough to make a grammar LL(1).
    - True
    - False

148. S --> A B A --> e | aA B --> e | bB - FIRST(S) contains ____ elements.
    - 3
    - 4
    - 5
    - 6

149. Grammars with LL(1) conflicts can be made LL(1) by applying left-factoring, substitution, and left-recursion removal. Left-factoring takes care of_____conflicts.
    - FIRST/FIRST
    - First/SECOND
    - SECOND/FIRST
    - NONE OF THE GIVEN

150. In an attribute grammar each production rule(N--> a) has a corresponding attribute evaluation rule that describes how to compute the values of the_____attributes of each particular node N in the AST.
    - Synthesized
    - Complete
    - Free
    - Bound

151. When constructing an LR(1) parser we record for each item exactly in which context it appears, which resolves many conflicts present in_____parsers based on FOLLOW sets.
    - SLR(1)
    - LRS(1_
    - RLS(1)

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- None of the Given

152. Backpatching to translate flow-of-control statements in _____ pass.

- one
- two
- three
- all of the given

153. LR parsing_____a string to the start symbol by inverting productions.

- Reduce
- Shift
- Adds
- None of the Given

154. _____ phase which supports macro substitution and conditional compilation.

- Semantic
- Syntax
- Preprocessing
- None of the Given

155. Parser always gives a tree like structure as output

- True
- False

156. Lexer and scanner are two different phases of compiler

- True
- False

157. _____tree in which each node represents an operator and children of the node represent the operands.

- Abstract Syntax
- Concrete Syntax
- Parse
- None of the Given

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158. Register allocation by graph coloring uses a register interference graph._____nodes in the graph are joined by an edge when the live ranges of the values they represent overlap.
- Two
- Three
- Four
- Five

159. In compilation process Hierarchical analysis is also called
- Parsing
- Syntax Analysis
- Both Parsing and Syntax analysis
- None Of the Given

160. Ambiguity can easily be handled by Top-down Parser
- True
- False

161. Front-end of a two pass compiler is consists of Scanner.
- True
- False

162. LL(1) parsing is called non-predictive parsing.
- True
- False

163. In predictive parsing table the rows are_____.
- Non-Terminal
- Terminals
- Both A and B
- None of the Given

164. In LL1() parsing algorithm_____contains a sequence of grammar symbols.
- Stack
- Link List
- Array

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- None of the Given

165. AST summarizes the grammatical structure with the details of derivations.
- True
- False

166. f X is a non-terminal in A? aX•?, then the interpretation of this transition is more complex because non-terminals do not appear in input
- Yes
- No

167. If / is a set of items for a grammar, then closure (/) is a set of items constructed from / by the following rule.

- If A --> aX.Y is in closure (/) and Y --> r is production, then add X --> .r to closure (/).
- If A --> a.XY is in closure (/) and X --> r is production, then add X --> .r to closure (/).
- If A --> aXY. is in closure (/) and A --> r is production, then add X --> .r to closure (/).
- None of these

168. NFA of LR(0) items means _____
- look-ahead one sybole
- no look-ahead
- look-ahead all sybols
- None of the given

169. A grammar is LR if a------- shift reduce-reduce parser can recognize handles when they appear on the top of stack.
- left-to-reverse
- left-to-rise
- left-to-right
- None of the given.

170. The output from the algorithm of constructing the collection of canonical sets of LR(1) items will be the _____
- Original Grammar G
- Augmented grammar G'
- Parsing table
- None of the given

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171. Reduction of a handle to the ------- on the left hand side of the grammar rule is a step along the reverse of a right most derivation.
- Terminal
- Non-terminal

172. NFA of LR(1) items means _____
- no look-ahead
- look-ahead one sybole
- look-ahead all sybols
- None of the given

173. performing common subexpression elimination on aa dependncy graph requires the identification of nodes with the same operator and operands.when using a hash table (with a hash function based on operator and operands) all_____nodes can be identified in linear time.
- common
- uncommon
- next
- previous

174. Linear IRs resemble pseudo-code for same _____.
- Automated Machine
- Mechanical machines
- Token machines
- Abstract machine

175. The regular expressions a*|b* and (a|b)* describe the_____set of strings.
- Same
- Different
- Onto

176. Back patching to translate flow-of-control statements in _____ pass.
- one Page no : 111
- two
- three
- all of the given

177. Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string “abbcde” can be parsed bottom-up by the following reduction steps: (i) aTbcde (ii) aTde (iii) aTUe (iv) S So

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what can be a handle from the following?

- The whole string, (aTUe)
- The whole string, (aTbcde)
- The whole string, (aTde)
- None of the given

178. Yacc contains built-in support for handling ambiguous grammars resulting in_____conflicts.

- Shift-reduce
- Shift-Shift
- Shift-second
- None of the given

179. The following two items A -> P • Q B -> P • Q can co-exist in an _____item set.

- LR
- LS
- LT
- PR

180. The error handling mechanism of the yacc parser generator pushes the input stream back when inserting 'missing' tokens.

- True
- False

181. Flow of values used to calculate synthesized attributes in the parse tree is:

- Bottom-up
- Right to left
- Top-Down
- Left to right

182. A lexical analyzer transforms a stream of tokens. The tokens are stored into symbol table for further processing by the parser.

- True
- False

183. LR parsers can handle _____ grammars.

- Left-recursive Page no: 163
- file-recursive
- End-recursive

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- Start-recursive

184. For each language to make LL(1) grammar, we take two steps, 1st is removing left recurrence and 2nd is applying fin sequence.

- True
- False

185. Can a DFA simulate NFA?

- Yes
- No
- Sometimes
- Depend upon nfa

186. Which of the statement is true about Regular Languages?

- Regular Languages are the most popular for specifying tokens.
- Regular Languages are based on simple and useful theory.
- Regular Languages are easy to understand.
- All of the given

187. The transition graph for an NFA that recognizes the language (a | b)*abb will have following set of states.

- {0}
- {0,1}
- {0,1,2}
- {0,1,2,3} not sure

188. Functions of Lexical analyzer are?

- Removing white space
- Removing constants, identifiers and keywords
- Removing comments
- All of the given

189. Consider the following grammar, S --> aTUe T --> Tbc/b U --> d And suppose that string “abbcde” can be parsed bottom-up by the following reduction steps: (i) aTbcde (ii) aTde (iii) aTUe (iv) S So, what can be a handle from the following?

- The whole string, (aTUe) Page no : 68
- The whole string, (aTbcde)
- The whole string, (aTde)
- None of the given

190. The LR(1) items are used as the states of a finite automaton (FA) that maintains information about the parsing stack and progress of a shift-reduce parser.

- True Page no: 74
- False

191. Flex is an automated tool that is used to get the minimized DFA (scanner).
    - True
    - False

192. We use ----- to mark the bottom of the stack and also the right end of the input when considering the Stack implementation of Shift-Reduce Parsing.
- Epsilon
- #
- $
- None of the given

193. When generating a lexical analyzer from a token description, the item sets (states) are constructed by two types of "moves": character moves and_____moves.
- E (empty string) Page no : 18
- #
- @
- none of given

194. Let a grammar G = (Vn, Vt, P, S) is modified by adding a unit production S'--> S to the grammar and now starting non-terminals becomes S' and grammar becomes G' = (Vn U {S'}, Vt, PU{S' --> S}, S'). The Grammar G' is called the ----------
- Augmented Grammar Page no : 76
- Lesser Grammar
- Anonymous Grammar
- none of given

195. Parser takes tokens from scanner and tries to generate _____.
    - Binary Search Tree
    - Parse Tree
    - Syntax Trace
    - None of the Given

196. In Flex specification file different sections are separated by___.
    - %% Page no: 26

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- &&
- ##
- \\

197. In DFA minimization we construct one_____for each group of states from the initial DFA.

- State Page no : 25
- NFA
- PDA
- None of Given

198. Intermediate Representation (IR) stores the value of its operand in _____.

- Registers
- Memory
- Hard Disk
- Secondary Storage

199. In_____certain checks are performed to ensure that components of a program fit together meaningfully.

- Linear analysis
- Hierarchical analysis
- Semantic analysis Page no : 33
- None

199. A_____is a top down parser.

- Predictive Parsing
- Reactive parser
- Proactive parser
- None of the given

200. Lexical Analyzer generator_____is written in Java.

- Flex
- Jlex Page no : 26
- Complex

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- None of given

201. _____avoid hardware stalls and interlocks.

- Register allocation
- Instruction scheduling
- Instruction selection
- None of given

202. Recursive _____parsing is done for LL(1) grammar.

- Decent
- Ascent
- Forward
- Backward

203. NFA of LR(1) items means___

- no look-ahead
- look-ahead one symbols
- look-ahead all symbols
- None

204. In the Parsing Table, the rows correspond to Parsing DFA states and columns correspond to ____

- Terminals and Non-terminals
- Start Symbol and its derivation
- Handles and derivations
- None

205. A grammar is LR if a___ shift reduce-reduce parser can recognize handles when they appear on the top of the stack

- left-to-reverse
- left-to-rise
- left-to-right
- None

206. Suppose ? begins with symbol X which may be a terminal (token) or non-terminal. The item can be written as A?Xa.?

- True
- False

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207. A handle is a substring that matches a___ side of production rule in the grammar.

- right hand
- left hand

208. If T->XYZ is a production of grammar G then which of the following item indicates that a string derivable from X has been seen so far on the input and we hope to see a string derivable from YZ next on the input.

- T->.XYZ
- T->X.YZ
- T->XY.Z
- T->XYZ.

209. In the canonical collection procedure, a DFA can not be constructed from NFA using the subset construction, similar to the one we used for lexical analysis.

- True
- False

210. Suppose ? begins with symbol X which may be a terminal (token) or non-terminal. The item can be written as ____

- A?a.X?
- A?Xa.?
- A?X?.
- X?Aa.?

211. If / is a set of items for grammar then closure(/) is a set of items constructed from / by the following rule.

- Every item in / is in closure(/) \
- Every item in / is not in closure(/)
- Only one item in / is in closure(/)
- None

212. NFA of LR(0) items means___

- no look ahead symbol

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- look ahead one symbol
- look ahead all symbols
- All of the given

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Compiler Construction (CS606)-

(Covid-19)

Solved by Sonu Ilyas Mughal

Sonu Mughal

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Question # 1 of 30 (Start time: 03:25:36 PM, 11 July 2020)

The input specification file to flex consists of _____sections:

Select the correct option

○	One
○	Two
■	Three
○	Four

BC140201349: Sidrah Afooj

CS606:Midterm Grand Quiz

Question # 2 of 30 (Start time: 03:26:33 PM, 11 July 2020)

Lexer and scanner are two different phases of compiler

Select the correct option

○	True
●	False

Question # 4 of 30 (Start time: 03:28:55 PM, 11 July 2020)

In compiler linear analysis is also called.

Select the correct option

○	Lexical analysis
○	Scanning
●	Both lexical analysis and scanning
○	None of the given

Question # 5 of 30 (Start time: 03:29:59 PM, 11 July 2020)

Which of the following statements is NOT true?

Select the correct option

○	The language accepted by finite automata are the languages denoted by regular expressions.
○	For every DFA there is a regular expression denoting its language.
●	For a regular expression r, there does not exist NFA with L(r) any transit that accept.
○	None of the given

Question # 6 of 30 (Start time: 03:31:12 PM, 11 July 2020)

A --> B C D
B --> h B | episilon
C --> C g | g | C h | i
D --> A B | episilon

First of A is _____ .

Select the correct option

●	h, g, i
○	g
○	h
○	None of the given.

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Question # 7 of 30 (Start time: 03:32:22 PM, 11 July 2020)

Top-down parsing expands a _____ from the start symbol to the leaves.

Select the correct option

- ○ Parse tree
- ○ Abstract Syntax tree
- ○ Parse tree
- ● All of the given

Question # 9 of 30 (Start time: 03:34:51 PM, 11 July 2020)

A top-down parser starts with the _____ of the parse tree.

Select the correct option

●	Root
○	Leaf
○	Middle
○	None of the given

Question # 10 of 30 (Start time: 03:35:17 PM, 11 July 2020)

Can a DFA simulate NFA?

Select the correct option

○	NO
●	YES
○	Sometimes
○	Depends on NFA

Question # 11 of 30 (Start time: 03:36:07 PM, 11 July 2020)

Bottom-up parsing is also called _____ .

Select the correct option

- (●) LR parsing
- ( ) LT parsing
- ( ) LS parsing
- ( ) SS parsing

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Question # 12 of 30 (Start time: 03:37:30 PM, 11 July 2020)

The _____ is optimized for hardware it is to run on.

Select the correct option

○	C++ code
○	C code
●	Assembly code
○	None of the given

Question # 13 of 30 (Start time: 03:38:34 PM, 11 July 2020)

Abstract syntax tree summarizes _____without the details of derivation.

Select the correct option

○	Theory
●	Grammatical structure
○	Data
○	None of the give

Question # 14 of 30 (Start time: 03:39:07 PM, 11 July 2020)

LR parsing _____ a string to the start symbol by inverting productions.

Select the correct option

- (●) Reduces
- ( ) Shifts
- ( ) Adds
- ( ) None of the given

Question # 15 of 30 (Start time: 03:39:43 PM, 11 July 2020)

Alternative to backtrack in top-down parser is _____.

Select the correct option

○	Context free grammar
○	Trees
●	Look ahead
○	None of the given

Question # 16 of 30 (Start time: 03:40:46 PM, 11 July 2020)

Typical compilation means programs written in high-level languages to low-level _____.

Select the correct option

●	Object code
○	Byted code
○	Unicode
○	Both Object Code and byte code

Question # 17 of 30 (Start time: 03:41:52 PM, 11 July 2020)

The _____ checks the stream of words (tokens) and their parts of speech for grammatical correctness.

Select the correct option

○	Parser
○	Scanner
●	Compiler
○	None of the given

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Question # 18 of 30 (Start time: 03:43:20 PM, 11 July 2020)

A grammar must be _____ before use for predictive parsing.

Select the correct option

○	Right factored
●	Left factored
○	Factored
○	None of the given

Question # 19 of 30 (Start time: 03:44:35 PM, 11 July 2020)

Parser does not distinguish between valid and invalid sequences of tokens.

Select the correct option

- ○ True
- ● False

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Question # 20 of 30 (Start time: 03:45:15 PM, 11 July 2020)

In compilation process Hierarchical analysis is also called

Select the correct option

○	Parsing
●	Syntax analysis
○	Both Parsing and Syntax analysis
○	None of given

Question # 21 of 30 (Start time: 03:46:22 PM, 11 July 2020)

The pair <role,word> is given the name_____.

Select the correct option

○	Word
●	Token
○	Syntax
○	None of the given

Question # 23 of 30 (Start time: 03:48:14 PM, 11 July 2020)

The back end maps Intermediate Representation (IR) into target _____.

Select the correct option

○	Object code
○	Machine code ✓
●	Source code
○	Linker

Question # 24 of 30 (Start time: 03:49:15 PM, 11 July 2020)

_____is evaluated to yield a value.

Select the correct option

- ○ Command
- ● Expression
- ○ Declaration
- ○ Declaration and Command

Question # 25 of 30 (Start time: 03:50:00 PM, 11 July 2020)

Front-end of a two pass compiler is consists of Scanner.

Select the correct option

- (●) True
- ( ) False

Question # 26 of 30 (Start time: 03:51:30 PM, 11 July 2020)

The _____returns a sequence of matching tokens at the output (or an error) and it always return the longest matching token.

Select the correct option

- ○ Scanner
- ○ Lexer
- ● Lexical analyzer
- ○ All of the given

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Question # 27 of 30 (Start time: 03:52:26 PM, 11 July 2020)

Parser takes tokens from scanner and tries to generate _____ .

Select the correct option

- ○ Binary Search tree
- ● Parse tree
- ○ Syntax trace
- ○ None of the given

Question # 28 of 30 (Start time: 03:53:02 PM, 11 July 2020)

_____avoid hardware stalls and interlocks.

Select the correct option

○	Register allocation
●	Instruction scheduling
○	Instruction selection
○	None of given

Question # 29 of 30 (Start time: 03:53:45 PM, 11 July 2020)

The handle-finding mechanism is the key to efficient _____parsing.

Select the correct option

○	Top-down
●	Bottom-up
○	Predictive
○	None of the given

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Question No : 1 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ convert the relocatable machine code into absolute machine code by linking library and relocatable object files.

Answer (Please select your correct option)

- Assembler
- Loader/link-editor

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- Compiler
- Preprocessor

Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Parsers take ____ as input from lexical analyzer.

Answer (Please select your correct option)

- Linker
- Token

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- Instruction
- None of the given

Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The regular expression _____denotes, the set of all strings of a's and b's of length two

Answer (Please select your correct option)

- a*
- (a*|b*)*
- (a*b*)*
- (a|b)(a|b)

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is a regular expression for the set of all strings over the alphabet {a} that has an even number of a's.

Answer (Please select your correct option)

- aa*
- (aa)*
- aa*a
- a(aa)*

check

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

Made by: Waqar Siddhu

Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ phase supports macro substitution and conditional compilation.

Answer (Please select your correct option)

- Semantic
- Syntax
- Preprocessing
- None of given

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

Made by: Waqar Siddhu

Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In LL(1) parsing algorithm, ____ contains a sequence of grammar symbols.

Answer (Please select your correct option)

- Stack

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- Link List
- Array
- None of the given.

Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the grammar

$$A \rightarrow B\ C\ D$$
$$B \rightarrow h\ B \mid \varepsilon$$
$$C \rightarrow C\ g \mid g \mid C\ h \mid i$$

Answer (Please select your correct option)

- h, g, i

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- g
- h
- None of the given.

Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ parsers never shifts into an error state.

Answer (Please select your correct option)

- LS
- LT
- LR
- LP

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

Question No : 9 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In parser, the two LL stand for _____ .

Answer (Please select your correct option)

- Left-to-right scan of input
- left-most derivation
- Left-to-right scan of input and left-most derivation

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- None of the given

Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ is elaborated to produce bindings.

Answer (Please select your correct option)

- Declaration

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- Expression
- Command
- None of the given

Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A lexical analyzer generated by ____ is essentially a FSA.

Answer (Please select your correct option)

- Dex
- Mex
- Fex
- Lex

Correct answer solved by Hadi
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Cell no: 03228043306

Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The actions (shift, reduce) in a SLR(1) parser depend on a lookahead symbol (_____) .

Answer (Please select your correct option)

- Current input token

Correct answer solved by Hadi
Email : usmanraj20@gmail.com
Cell no: 03228043306

- Next Input Token
- Previous output Token
- Previous Input Token.

Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The following grammar contains a _____ conflict.

$S \rightarrow A \mid xb$

Answer (Please select your correct option)

- Shift-Reduce

Correct answer solved by Hadi
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- First-Reduce
- Shift-First
- Reduce-Reduce

Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Considering the following grammar:

$S \rightarrow AB$
$A \rightarrow \varepsilon \mid aA$
$B \rightarrow b \mid bB$

Answer (Please select your correct option)

- 1
- 2
- 3
- 4

Question No : 15 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Simple code generation considers one AST node at the time. When the target is a *stack* machine, the code can be generated in one _____ traversal of the AST.

Answer (Please select your correct option)

- Depth-first

Correct answer solved by Hadi
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- Breadth-first
- First-first
- Shift-Reduce

Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ is a register allocation technique that *always* finds the minimal number of registers needed for a procedure.

Answer (Please select your correct option)

- Dangling reference
- Graph coloring

Correct answer solved by Hadi
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Cell no: 03228043306

- Left Factoring
- Right Recursion

Question No : 17 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When generating code at the basic block level, the dependency graph must be converted to target code. By identifying _____ , instruction selection and instruction ordering can be performed efficiently in a single pass.

Answer (Please select your correct option)

- Token sequences
- Ladder sequences
- Physical sequences
- Logical sequences

check

Correct answer solved by Hadi
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Cell no: 03228043306

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ can be considered a small compiler since it transforms a source language (assembly) into a less abstract target language (binary object code).

Answer (Please select your correct option)

- Parser
- Assembler
- Lexical analyzer
- Scanner

Correct answer solved by Hadi
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Question No : 19 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When memory allocator operates on chunks which include some administrative part and a block of user data. The administrative part includes _____ flag for marking the chunk as free or in-use.

Answer (Please select your correct option)

- One

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- Two
- Three
- Four

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

A parser transforms a stream of tokens into an _____ .

Answer (Please select your correct option)

- AST

Correct answer solved by Hadi
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- IST
- EST
- ATS

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Question No : 21 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The parser generator yacc can handle _____ grammars.

Answer (Please select your correct option)

- LL(1)
- LT(1)
- LS(1)
- LF(1)

Correct answer solved by Hadi
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Question No : 22 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Simple code generation considers one AST node at a time. If the target is a *register* machine, the code can be generated in one _____ traversal of the AST, possibly introducing temporaries when running out of registers.

Answer (Please select your correct option)

- Depth-first

Correct answer solved by Hadi
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- Breadth-first
- Depth-second
- Breadth-second

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Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Attributes whose values are defined in terms of a node's own attributes, node's siblings and node's parent are called _____ .

Answer (Please select your correct option)

- Inherited attributes

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- Physical attributes
- Logical attributes
- Un-synthesized attributes

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

A linker combines multiple object files into a _____ executable object.

Answer (Please select your correct option)

- Single
- Double
- Triple
- Quadruple

Correct answer solved by Hadi
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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

The notation _____ instructs YACC to push a computed attribute value on the stack.

Answer (Please select your correct option)

- $$
- &&
- ##
- --

Correct answer solved by Hadi
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Question No : 26 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The following two items

$A \rightarrow P \cdot Q$

$B \rightarrow P \cdot Q$

can co-exist in an _____ item set.

Answer (Please select your correct option)

- LR

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- LS
- LT
- PR

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Question No : 26 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The following two items

$A \rightarrow P \bullet Q$

$B \rightarrow P \bullet Q$

can co-exist in an _____ item set.

Answer (Please select your correct option)

- LR

Correct answer solved by Hadi
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- LS
- LT
- PR

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Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When generating a lexical analyzer from a _____ description, the item sets (states) are constructed by two types of "moves": character moves and ε moves.

Answer (Please select your correct option)

- Character
- Grammar
- Token
- Sentence

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Hybrid IRs combine elements of _____ .

Answer (Please select your correct option)

- Graphical (structural)
- Linear IRs
- Both graphical and linear IRs
- Non-Linear IRs

Correct answer solved by Hadi
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Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

x[i] = y

This is _____ .

Answer (Please select your correct option)

- Prefix assignment
- Postfix assignment
- Index assignment
- Non-Index assignment

Correct answer solved by Hadi
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Question No : 30 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A lexical analyzer generator, automatically constructs a _____ that recognizes tokens.

Answer (Please select your correct option)

- FA
- PDA
- DP
- Unidirectional Graph

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Question No : 31 of 52 **Marks: 1 (Budgeted Time 1 Min)**

if x relop y goto L

Above statement is _____

Answer (Please select your correct option)

- Abstract jump
- Conditional jump
- While loop
- Unconditional jump

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

In a CFG (Context Free Grammar) the set of terminal and non-terminal symbols must be _____.

Answer (Please select your correct option)

- Disjoint

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- Logical
- Relational
- Joint

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Question No : 33 of 52 **Marks: 1 (Budgeted Time 1 Min)**

YACC contains built-in support for handling ambiguous grammars resulting in _____ conflicts.

Answer (Please select your correct option)

- Shift-reduce

Correct answer solved by Hadi
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- Shift-Shift
- Reduce-reduce
- Reduce-Shift

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Question No : 34 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The _____ translation statements can be conveniently specified in YACC.

Answer (Please select your correct option)

- Syntax-directed

Correct answer solved by Hadi
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- Image-directed
- Sign-directed
- Segment-directed

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Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

When constructing an LR(1) parser we record for each item exactly in which context it appears, which resolves many conflicts present in _____ parsers based on FOLLOW sets.

Answer (Please select your correct option)

- SLR(1)
- LRS(1)
- RLS(1)
- SLL(1)

Correct answer solved by Hadi
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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Code generation module has to tackle _____.

Answer (Please select your correct option)

- Memory management
- Instruction selection
- Instruction scheduling
- All of the given

Correct answer solved by Hadi
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Question No : 37 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For convenience, lexical analyzers should read the complete _____ program into memory.

Answer (Please select your correct option)

- Input
- Output
- Input and output
- Tokens

Correct answer solved by Hadi
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Question No : 38 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Considering the following grammar:

S → A | x
A → aAb | x

The grammar contains a _____ conflict.

Answer (Please select your correct option)

- Reduce-reduce

Correct answer solved by Hadi
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- First-first
- Shift-shift
- Shift-reduce

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The order in which the DAG is traversed can lead to _____ code

Answer (Please select your correct option)

- Better

Correct answer solved by Hadi
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- Worse
- Large
- Garbage

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Register allocation problem uses the strategy of _____ .

Answer (Please select your correct option)

- Graph coloring

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- Graph nodding
- Graph edging
- Graph patching

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